

Contrasting conservation conditions shape the spatial partitioning of zooplankton diversity in Atlantic Forest streams

Contrasting conservation conditions shape the spatial partitioning of zooplankton diversity in Atlantic Forest streams ON Acta Limnologica Brasiliensia

Ellen Gomes da Silva Mayara Mirelly da Silva Monteiro
Marcela Vitória Bernardo da Silva
Ludmilla Cavalcanti Antunes Lucena Thais Xavier de Melo
Elvio Sergio Figueredo Medeiros

2026-09-30

Abstract

Table of contents

1	Introduction	2
1.1	Objectives	3
2	Material and methods	3
2.1	Study area and sampling design	3
2.2	Environmental data collection	4
2.3	Zooplankton sampling	5
2.4	Statistical analyses	6
3	Results	17
3.1	Environmental variables and nutrient concentration	17
3.2	Community structure	36
3.3	Diversity Partitioning	50
4	Discussion	61
	Conclusions	61

Acknowledgements	61
Authorship contribution statement	61
Ethical approval	61
Declaration of generative AI and AI-assisted technologies in the manuscript preparation process	62
References	62
Figures and Tables	65
Figures	66
Tables	67
Apendices	76
Non-used figures and tables	76

List of Figures

1	66
2	67
3	68
4	69
5	70
6	71
7	72

List of Tables

1 Environment data GT table.	73
2 Nutrient concentrations at the sampling sites.	73
3 Density (ind/m ²) taxa GT table.	74
4 Partition data GT table.	75

1 Introduction

Essas abordagens são baseadas nos conceitos de turnover and nestedness. Species turnover and nestedness are two distinct components used to break down and understand beta diversity—

which measures how much species composition changes between different communities or sites (Baselga, 2010), where, turnover is the replacement of some species by different species from one site to another, while nestedness is the loss or gain of species where poorer sites contain a strict subset of the species found at richer sites.

CITAR: (Ávila et al., 2011; Baselga, 2010, 2012; Crist et al., 2003; Ligeiro et al., 2010; Shell & Collier, 2018)

1.1 Objectives

This study describes the richness and density of zooplankton, comparatively between two small head water streams subject to different levels of environmental pressures. One originates within a protected area - Biological Reserve, or REBIO, according to Brasil (2000) - and its course extends beyond its administrative boundaries (MMA, 2003), and a second one that flows entirely outside of the protected area. Within this framework, we use hierarchical additive partitioning to evaluate the relative contribution of different spatial and temporal scales to the diversity of zooplankton assemblages. Given the importance of the Atlantic Forest to the biodiversity (Pereira et al., 2026) and its fragmented stated north of the São Francisco river basin (Silva & Casteleti, 2003), we aim with this study, (1) to infer environmental degradation from species turnover (Baselga, 2010) and (2) to associate beta-diversity to the ecological health of sites inside and surrounding a protected area of Atlantic Forest. Given the expected different levels of environmental pressures in the area (Gouveia et al., 2017), we hypothesize that the diversity is not be homogeneously distributed among the different spatial scales but concentrated at the finer levels and seasonal variation will be limited to species composition, but no spacial variation is expected between dry and wet seasons. Some environmental features (reach morphology, habitat structure, and water quality) are also described to explore possible explaining factors for the observed patterns.

2 Material and methods

2.1 Study area and sampling design

This study was conducted in the Guaribas Biological Reserve (REBIO Guaribas), a protected area of Atlantic Forest in northeastern Brazil. This protected area is located in a transitional zone between the Caatinga and Atlantic Forest ecoregions. The land use of the study area is based on the occupation of traditional human populations and an increasing use of land for large scale monocultures, mostly sugar cane (Heinrichs et al., 2017). Other than these, small scale agricultural activities, fishing, and the modification of the natural vegetation for livestock are important anthropogenic pressures that affect the buffer zone of the REBIO (Arruda et al., 2013; MMA, 2026). Other common uses of the soil in the area are, coconut plantations, orchards and subsistence agriculture. These land uses represent important sources

of pollution, either through agrochemicals runoff or direct and indirect domestic effluents (such as septic tanks). These nonpoint source pollutants contaminate underground water and create a spatial gradient of environmental disturbance (Mehla et al., 2026). Therefore, the mainly rural nature of the land use associated with the management of vegetation, will cause direct impact to aquatic systems, by the siltation of streams and contamination of groundwater. The hydrographic network of the study region is composed predominantly of small catchments containing first- and second-order streams (Strahler, 1957 classification) (Figure 1). These watercourses are tributaries of the Camaratuba River and are sustained by numerous springs distributed throughout the region. The two main stream systems in the study area are the Barro Branco and Caiana catchments. The Barro Branco stream is approximately 8.8 km long and originates within the protected area. Its upper section remains within the boundaries of the REBIO, whereas its middle and lower sections extend into the surrounding landscape. Despite this transition, riparian vegetation is preserved along most of the stream. The Caiana stream is approximately 9.8 km long and also originates within the REBIO, although most of its course lies outside the protection area.

The climate in the region is classified as As' (Peel et al., 2007), characterized by a mean annual temperature of approximately 25 °C, with maximum temperatures occurring mainly between February and March, during the dry period, as well as by low annual thermal amplitude. Annual precipitation ranges from 1,400 to 1,800 mm, concentrated between March and July, whereas the lowest rainfall levels occur between September and December. In the study area, rainfall is concentrated between March and July, beginning in autumn and extending into winter. This pattern is attributed to the influence of cold air masses from the South Atlantic ocean that affect the region. The hydrographic network is composed of small headwater streams fed by rain (Gouveia et al., 2017).

Sampling was conducted during the dry (January-February) and rainy (June-July) periods of 2024 in two Atlantic Forest streams. A total of 15 sampling sites were distributed longitudinally across the two streams to capture spatial variation in environmental conditions and habitat characteristics. The Barro Branco stream included sampling sites located within the protected area and sites located outside its boundaries, whereas all sampling sites in the Caiana stream were located outside the protected area. This sampling design allowed the assessment of zooplankton diversity across nested spatial scales and contrasting conditions of environmental protection and human influence (Figure 1).

2.2 Environmental data collection

To evaluate whether the a priori classification of sampling sites into Types I–IV was supported by quantitative environmental information, environmental conditions were characterized using four groups of variables: site morphology, water quality, substrate composition, and marginal habitat structure. Morphological characteristics were assessed from three randomly selected transects at each site, where stream width (cm) and water depth (cm) were measured. Bank slope was classified into four categories according to its inclination: 1, < 20°; 2, > 20–30°; 3, >

30–50°; and 4, > 50°. Broader-scale landscape characteristics, including elevation and stream length, were obtained from handheld GPS records and satellite imagery. Water quality was characterized using physical and chemical measurements obtained with portable instruments, including water temperature (°C) and dissolved oxygen concentration (mg/L). Turbidity was subsequently measured in the laboratory with a turbidimeter using water samples collected at each sampling site. Current velocity (m/s) was estimated using the float method, as described by Maitland (1990). Substrate composition and the physical structure of marginal habitats were evaluated following procedures adapted by Medeiros et al. (2008) from the protocols proposed by Pusey et al. (2004) and Mugodo et al. (2006). Between 9 and 12 quadrats (1 m²) were distributed along the terrestrial–aquatic interface at each sampling site. Within each quadrat, the proportional cover of different substrate categories was visually estimated, including mud, sand, cobbles, small and large gravel and rocks. The same quadrats were used to characterize littoral and submerged habitat features. Recorded habitat components included macrophytes, marginal grasses, leaf litter, attached and filamentous algae, overhanging and submerged vegetation, small and large woody debris, and root masses. For subsequent analyses and presentation, environmental measurements were summarized as mean values for each sampling site.

Three 250 mL replicate samples of water from each site were taken and refrigerated for concentration analysis (g L⁻¹) of ammonia, nitrite, nitrate, orthophosphate (soluble reactive phosphorus) and total phosphorus. Nutrient concentrations were obtained based on Standard Methods for the Examination of Water (1998) and Mackereth et al. (1978).

The ecophysiological evaluation of human impact on the forest, was mainly guided by the fish species inventory for the REBIO made by Gouveia et al. (2017). This study lists 18 species from five different orders: Characiformes (12 species), Perciformes (3 species), Synbranchiformes (1 species), Siluriformes (1 species) e Cyprinodontiformes (1 species). From those, two are exotic - *Oreochromis niloticus*, second most farmed fish species in the world and tolerant to osmotic variation, and *Poecilia reticulata*, largely used in mosquito control and resistant to stressful environments (Miranda, 2012). The predominance of small fish inhabiting small basins is thought to lead to low dispersion and consequent isolation of populations (Gouveia et al., 2017), associated with the fact that most species in the study area are phylogenetic close (with the predominance of Characidae) and are widely distributed, offers an appropriate design to access the entire local ichthyofauna using a few species, namely the genres *Hemigrammus* and *Astyanax*.

2.3 Zooplankton sampling

Zooplankton was sampled at habitats within each sampling site, hereafter referred to as habitat samples. Collections were performed in the main channel of the streams at the surface water. At each sampling habitat, a plankton net (30 cm mouth diameter, 70 cm length, and 60 µm mesh size) was used to filter 50 L of water. The net was rinsed between successive samples to prevent clogging and carryover of organisms. Each sample was concentrated to a final

volume of 80 mL for laboratory analyses. Following collection, zooplankton samples were anesthetized with commercial sparkling water and subsequently preserved in 4% formalin. Sucrose was added to the preserved material to reduce the loss of eggs by female cladocerans and minimize deformation of zooplankton carapaces (Haney & Hall, 1973). In the laboratory, samples had their zooplankton identified and counted using a Sedgewick–Rafter counting chamber. The analyses focused on representatives of Rotifera, Cladocera, and Copepoda. Taxonomic identification was based on specialized literature and confirmed by specialists. To evaluate whether the sampling effort adequately represented the zooplankton assemblage, species accumulation curves were calculated

Zooplankton was collected using a plankton net (opening diameter 30 cm, 70 cm long, pore size 60 µm and sample cup of 80 mL). At three different locations, at each site, one sample of fifty liters of water was filtered through the net. Commercial sparkling water was added to the sample to narcotize the zooplankters, which were then preserved in 4% formalin (EPA 2013). Sucrose was added to the preserved sample to prevent female cladocerans from losing eggs and to minimize carapace distortion (Haney e Hall 1973). In the laboratory, three subsample aliquots of 1 mL were taken from each sample and all individuals were identified and counted in a Sedgewick-Rafter counting cell (Koste 1978, Elmoor-Loureiro 1997, Elmoor-Loureiro e Sousa 2021, GBIF.org 2023, NIES 2023, Haney J. F. et al 2024). Only rotifers, cladocerans and copepods were considered in the present study.

2.4 Statistical analyses

Environmental variables were screened for multicollinearity and to identify redundant descriptors. Variables representing similar environmental information were also combined when appropriate, thereby reducing redundancy in the dataset (Legendre & Legendre, 1998).

```

1  #ZOOEBIO23_PART - ORGANIZANDO DADOS AMBIENTAIS----
2  #####.....-----
3
4  dev.off() #apaga os graficos, se houver algum
5  rm(list=ls(all=TRUE)) #limpa a memória
6  cat("\014") #limpa o console
7
8  ##CARREGANDO MATRIZES BRUTAS-----
9
10 library(openxlsx)
11 habitat <- read.xlsx("dados/rebio23-habitat.xlsx",
12                      rowNames = T,
13                      colNames = T,
14                      sheet = "ambiente",
15                      startRow = 2)
16 # Identify columns containing only NA values

```

```

17 all.na <- names(habitat)[colSums(!is.na(habitat)) == 0]
18 all.na <- c(all.na, "w.Temp_C", "w.DO_mg.L")
19 # Remove those columns by name
20 m_hab_part <- habitat[, !names(habitat) %in% all.na, drop = FALSE]
21 # Removendo linhas por nome
22 rownames(m_hab_part)
23 del_rows <- c("C-C-P11-H1", "C-C-P11-H2", "C-C-P11-H3",
24             "S-C-P11-H1", "S-C-P11-H2", "S-C-P11-H3")
25 del_rows
26 m_hab_part <- m_hab_part[!(row.names(m_hab_part) %in% c(del_rows)),]
27 m_hab_part
28 # Remove columns with at least one NA
29 m_hab_part <- m_hab_part[, colSums(is.na(m_hab_part)) == 0, drop = FALSE]
30
31 ###REMOVENDO COLUNAS ZERADAS DE HABITAT----
32
33 sum <- colSums(m_hab_part)
34 sum
35 zero_sum <- names(which(colSums(m_hab_part) == 0))
36 zero_sum #nomes das colunas zeradas
37 m_part_cols <- m_hab_part[(colSums(m_hab_part) != 0)] #em != a exclamação inverte o sentido
38 zero_sum2 <- names(which(colSums(m_part_cols) == 0))
39 zero_sum2 #nomes das colunas zeradas
40 sum<-colSums(m_part_cols)
41 sum
42
43 t_grps <- read.xlsx("dados/rebio23-habitat.xlsx",
44                   rowNames = T,
45                   colNames = T,
46                   sheet = "grupos",
47                   startRow = 2)
48
49 ###REMOVENDO LINHAS ZERADAS DE GRUPOS----
50
51 rownames(t_grps)
52 del_rows
53 t_grps <- t_grps[!(row.names(t_grps) %in% c(del_rows)),]
54 t_grps
55
56 ###SEPARANDO NUTRIENTES----
57
58 nutri <- grep("^n\\.\\.", colnames(habitat), value = TRUE)
59 nutri

```

```

60 nutrientes <- habitat[, nutri, drop = FALSE]
61 nutrientes <- nutrientes[, colSums(!is.na(nutrientes)) > 0, drop = FALSE]
62 nutrientes <- na.omit(nutrientes)
63
64 ##SALVANDO MATRIZES FINAIS-----
65
66 write.table(m_part_cols, "res/m_hab_part.csv",
67             sep = ";", dec = ".", #"\t",
68             row.names = TRUE,
69             quote = TRUE,
70             append = FALSE)
71 write.table(nutrientes, "res/t_nutri.csv",
72             sep = ";", dec = ".", #"\t",
73             row.names = TRUE,
74             quote = TRUE,
75             append = FALSE)
76 write.table(t_grps, "res/t_grps.csv",
77             sep = ";", dec = ".", #"\t",
78             row.names = TRUE,
79             quote = TRUE,
80             append = FALSE)
81 t_grps <- read.csv("res/t_grps.csv",
82                  sep = ";", dec = ".",
83                  row.names = 1,
84                  header = TRUE,
85                  na.strings = NA)
86 t_nutri <- read.csv("res/t_nutri.csv",
87                   sep = ";", dec = ".",
88                   row.names = 1,
89                   header = TRUE,
90                   na.strings = NA)
91 m_hab <- read.csv("res/m_hab_part.csv",
92                 sep = ";", dec = ".",
93                 row.names = 1,
94                 header = TRUE,
95                 na.strings = NA)
96
97 ##ZOOEBIO23_PART - ORGANIZANDO DADOS DA COMUNIDADE-----
98 #####....-----
99
100 dev.off() #apaga os graficos, se houver algum
101 rm(list=ls(all=TRUE)) #limpa a memória
102 cat("\014") #limpa o console

```



```

103
104 library(openxlsx)
105
106 ##CARREGANDO MATRIZES BRUTAS----
107
108 densidade <- read.xlsx("dados/rebio23-zoopl.xlsx",
109                       rowNames = T,
110                       colNames = T,
111                       sheet = "densidade_hab_indL")
112 densidade[1:5,1:5] #[1:5,1:5] mostra apenas as linhas e colunas de 1 a 5.
113
114 ##COLUNAS ZERADAS DE DENSIDADE----
115 sum <- colSums(densidade)
116 sum
117 zero_sum <- names(which(colSums(densidade) == 0))
118 zero_sum #nomes das espécies zeradas
119 m_part_cols <- densidade[(colSums(densidade) != 0)] #em != a exclamação inverte o sentido
120 zero_sum2 <- names(which(colSums(m_part_cols) == 0))
121 zero_sum2 #nomes das espécies zeradas
122 sum<-colSums(m_part_cols)
123 sum
124
125 ###LINHAS ZERADAS DE DENSIDADE----
126 m_part2 <- as.data.frame(t(m_part_cols))
127 sum <- colSums(m_part2)
128 sum
129 zero_sum <- names(which(colSums(m_part2) == 0))
130 zero_sum #nomes das espécies zeradas
131 m_part_rows <- m_part2[(colSums(m_part2) != 0)] #em != a exclamação inverte o sentido
132 zero_sum2 <- names(which(colSums(m_part_rows) == 0))
133 zero_sum2 #nomes das espécies zeradas
134 sum<-colSums(m_part_rows)
135 sum
136 m_dens <- as.data.frame((t(m_part_rows)))
137
138 ##SALVANDO MATRIZES FINAIS REMOVIDAS AS LINHAS/COLUNAS ZERADAS----
139
140 write.table(m_dens, "res/m_dens.csv",
141            sep = ";", dec = ".", #"\\t",
142            row.names = TRUE,
143            quote = TRUE,
144            append = FALSE)
145 m_dens <- read.csv(res/"m_dens.csv",

```

```

146         sep = ";", dec = ".",
147         row.names = 1,
148         header = TRUE,
149         na.strings = NA)
150
151 #ZOOOREBIO23_PART - MATRIZ DE CONTAGEM----
152 #####....----
153
154 library(openxlsx)
155 contagem <- read.xlsx("dados/rebio23-zoopl.xlsx",
156                      rowNames = T,
157                      colNames = T,
158                      sheet = "rebio23_80ml",
159                      startRow = 2)
160 contagem[1:5,1:5] #[1:5,1:5] mostra apenas as linhas e colunas de 1 a 5.
161
162 ###COLUNAS ZERADAS DE CONTAGEM----
163 sum <- colSums(contagem)
164 sum
165 zero_sum <- names(which(colSums(contagem) == 0))
166 zero_sum #nomes das espécies zeradas
167 m_part_cols <- contagem[(colSums(contagem) != 0)] #em != a exclamação inverte o sentido
168 zero_sum2 <- names(which(colSums(m_part_cols) == 0))
169 zero_sum2 #nomes das espécies zeradas
170 sum<-colSums(m_part_cols)
171 sum
172
173 ###LINHAS ZERADAS DE CONTAGEM----
174 m_part2 <- as.data.frame(t(m_part_cols))
175 sum <- colSums(m_part2)
176 sum
177 zero_sum <- names(which(colSums(m_part2) == 0))
178 zero_sum #nomes das espécies zeradas
179 m_part_rows <- m_part2[(colSums(m_part2) != 0)] #em != a exclamação inverte o sentido
180 zero_sum2 <- names(which(colSums(m_part_rows) == 0))
181 zero_sum2 #nomes das espécies zeradas
182 sum<-colSums(m_part_rows)
183 sum
184 m_cont <- (t(m_part_rows))
185 m_cont
186 sum(m_cont)
187 ncol(m_cont)
188

```

```

189  ##SALVANDO MATRIZ FINAL DE CONTAGEM----
190
191  write.table(m_cont, "res/m_cont.csv",
192             sep = ";", dec = ".", #"\\t",
193             row.names = TRUE,
194             quote = TRUE,
195             append = FALSE)
196  m_cont <- read.csv("res/m_cont.csv",
197                   sep = ";", dec = ".",
198                   row.names = 1,
199                   header = TRUE,
200                   na.strings = NA)
201  m_cont
202
203  #ZOOEBIO23_PART - ORGANIZANDO ESCALAS DA COMUNIDADE----
204  #####....----
205
206  dev.off() #apaga os graficos, se houver algum
207  rm(list=ls(all=TRUE)) #limpa a memória
208  cat("\\014") #limpa o console
209
210  library(openxlsx)
211
212  ##CARREGANDO MATRIZES BRUTAS----
213
214  escalas <- read.xlsx("dados/rebio23-zoopl.xlsx",
215                     rowNames = T,
216                     colNames = T,
217                     sheet = "escalas_final",
218                     startRow = 2)
219
220  rownames(escalas)
221  del_rows <- c("C-C-P11-H1", "C-C-P11-H2", "C-C-P11-H3", "S-C-P11-H1", "S-C-P11-H2", "S-C-P11-H3")
222  del_rows
223  escalas <- escalas[!(row.names(escalas) %in% c(del_rows)),]
224
225  write.table(escalas, "res/m_esc.csv",
226             sep = ";", dec = ".", #"\\t",
227             row.names = TRUE,
228             quote = TRUE,
229             append = FALSE)
230  m_esc <- read.csv("res/m_esc.csv",
231                   sep = ";", dec = ".",

```

```

232         row.names = 1,
233         header = TRUE,
234         na.strings = NA)
235 m_esc

1  #REBIO23-part----
2
3  ##ORGANIZANDO DADOS----
4
5  dev.off()
6  rm(list=ls(all=TRUE))
7  cat("\014")
8
9  t_grps <- read.csv("res/t_grps.csv",
10                  sep = ";", dec = ".",
11                  row.names = 1,
12                  header = TRUE,
13                  na.strings = NA)
14  m_hab <- read.csv("res/m_hab_part.csv",
15                  sep = ";", dec = ".",
16                  row.names = 1,
17                  header = TRUE,
18                  na.strings = NA)
19
20  ##CORRELOGRAMA E REMOÇÃO DE VARIÁVEIS REDUNDANTES OU DESNECESSÁRIAS----
21
22  library(psych)
23  colnames(m_hab)
24  png("res/fig-h.hab_pairs.png")
25  pairs.panels(m_hab[,1:8],
26              method = "pearson", # correlation method
27              scale = FALSE, lm = FALSE,
28              hist.col = "#00AFBB", pch = 19,
29              density = TRUE, # show density plots
30              ellipses = TRUE, # show correlation ellipses
31              alpha = 0.5)
32  dev.off()
33
34  cor <- cor(m_hab)
35  cor
36
37  library(corrplot)
38  png("res/fig-hab_corrplot.png")

```

```

39 corrplot(cor, method = "circle")
40 dev.off()
41
42 ##DELETANDO COLINEARES OU INDESEJADAS----
43
44 sink(file = "res/colineares.txt", append = F, split = T)
45 colnames(m_hab)
46 del_cols <- c("", "")
47 #del_cols <- grep("^n\\.\"", colnames(m_hab), value = TRUE)
48 m_hab_part <- m_hab[, !(colnames(m_hab) %in% del_cols)]
49
50 ##SOMANDO REDUNDANTES----
51
52 m_hab_part$h.Algae <- m_hab_part$h.Algae_f + m_hab_part$h.Algae_a
53 m_hab_part <- m_hab_part[, !(colnames(m_hab_part)
54                               %in% c("h.Algae_f", "h.Algae_a"))]
55 m_hab_part$h.Debris <- m_hab_part$h.Debris_l + m_hab_part$h.Debris_s
56 m_hab_part <- m_hab_part[, !(colnames(m_hab_part)
57                               %in% c("h.Debris_l", "h.Debris_s"))]
58
59 colnames(m_hab_part)
60 m_hab_part
61 sink()
62
63 write.table(m_hab_part, "res/m_hab_part.csv",
64             sep = ";", dec = ".", #"\\t",
65             row.names = TRUE,
66             quote = TRUE,
67             append = FALSE)
68 m_hab_part <- read.csv("res/m_hab_part.csv",
69                       sep = ";", dec = ".",
70                       row.names = 1,
71                       header = TRUE,
72                       na.strings = NA)
73
74 colnames(m_hab_part)
75 library(PerformanceAnalytics)
76 chart.Correlation(m_hab_part[, c(1:20)], histogram=TRUE)
77 # ***p<0.001, **p<0.01, *p<0.05, .p<0.10
78
79 library(Hmisc)
80 mat <- as.matrix(m_hab_part[, 1:20])
81 res <- rcorr(mat, type = "pearson")

```

```

9 cor_mat <- res$r
10 p_mat <- res$P
11
12 stars <- ifelse(p_mat < 0.001, "***",
13               ifelse(p_mat < 0.01, "**",
14                     ifelse(p_mat < 0.05, "*", "")))
15
16 cor_star <- ifelse(p_mat < 0.05,
17                  paste0(round(cor_mat, 2), stars),
18                  NA)
19
20 cor_star <- matrix(
21   cor_star,
22   nrow = nrow(cor_mat),
23   dimnames = dimnames(cor_mat)
24 )
25
26 cor_star[upper.tri(cor_star, diag = TRUE)] <- NA
27 cor_star <- noquote(cor_star)
28 print(cor_star, na.print = "")

```

These data were then analyzed using Principal Component Analysis (PCA) to summarize multivariate patterns of environmental variation among sampling sites. Prior to PCA, site morphology and water-quality variables were square-root transformed (Sokal & Rohlf, 1995). Variables describing sediment composition and marginal habitat structure, expressed as percentages, were relativized by their column totals and subsequently subjected to the arcsine square-root transformation (McCune & Grace, 2002). PCA was conducted in R using the FactoMineR package (Lê et al., 2008), with all variables centered and standardized to unit variance before analysis. The water quality parameters temperature and dissolved oxygen, and all nutrients were used to characterize differences between streams but were not used in the PCA because they were not measured for both seasons.

```

1  #PCA fviz package----
2  #browseURL("https://www.sthda.com/english/articles/31-principal-component-methods-in-r-pract
3
4  #ORGANIZANDO DADOS----
5
6  dev.off()
7  rm(list=ls(all=TRUE))
8  cat("\014")
9
10 t_grps <- read.csv("res/t_grps.csv",
11                  sep = ";", dec = ".",

```

```

12         row.names = 1,
13         header = TRUE,
14         na.strings = NA)
15 m_hab_part <- read.csv("res/m_hab_part.csv",
16                       sep = ";", dec = ".",
17                       row.names = 1,
18                       header = TRUE,
19                       na.strings = NA)
20
21 colnames(m_hab_part)
22
23 ###RELATIVIZAÇÕES E TRANSFORMAÇÕES----
24 library(vegan)
25 library(tidyverse)
26
27 # Selecionando variáveis que começam com "w" ou "m"
28 m_wm <- m_hab_part[, grep("^ (w|m)", colnames(m_hab_part))]
29
30 # Selecionando variáveis que começam com "s" ou "h"
31 m_sh <- m_hab_part[, grep("^ (s|h)", colnames(m_hab_part))]
32
33 # Removendo os dois primeiros caracteres dos nomes das colunas
34 colnames(m_wm) <- sub("^..", "", colnames(m_wm))
35 colnames(m_sh) <- sub("^..", "", colnames(m_sh))
36
37 # Transformações
38 m_wm <- sqrt(m_wm)
39 m_sh <- asin(sqrt(decostand(m_sh, method = "total", MARGIN = 2)))
40
41 # Juntando tudo em uma única tabela
42 hab_trns <- cbind(m_sh, m_wm)
43
44 # Salvado m_hab_trns
45 write.table(hab_trns, "res/m_hab_trns.csv",
46            sep = ";", dec = ".", #"\t",
47            row.names = TRUE,
48            quote = TRUE,
49            append = FALSE)
50 m_hab_trns <- read.table("res/m_hab_trns.csv",
51                        sep = ";", dec = ".",
52                        row.names = 1,
53                        header = TRUE,
54                        na.strings = NA)

```

Zooplankton diversity was characterized using species richness (R) and Shannon’s density (D) (Magurran, 2013). Species richness was calculated as the number of species recorded in each sample, whereas density was expressed as the number of individuals per liter (ind./L) (Bowen, 2017; Silva et al., 2017). Variation in richness and density was evaluated using factorial analysis of variance (ANOVA) followed by *post hoc* multiple comparisons using Tukey’s HSD test ($\alpha = 0.05$) (Zar, 2010). Density was square root transformed to enhance normality and homogeneity of variances (Sokal & Rohlf, 1995) and species richness was corrected based on rarefied analyses for the average number of individuals from all sites (Chao et al., 2014; Oksanen et al., n.d.).

Each observation was classified according to three factors: river reach (three levels), habitat (three levels), and hydrological season (wet and dry). The analysis initially considered the full factorial model. When significant interactions were detected, higher-order interactions were examined first, followed by the corresponding two-way interactions (reach $\times$ season, reach $\times$ habitat, and season $\times$ habitat) and, subsequently, the main effects. Response variables were transformed using $\log(x + 1)$ to improve compliance with assumptions of normality and homogeneity of variances [39]. Differences among factor levels were subsequently investigated using Tukey’s HSD test at $\alpha = 0.05$. $\rightarrow$

Differences in assemblage composition across the spatial hierarchy were explored using hierarchical agglomerative clustering. Bray–Curtis dissimilarities were calculated from the density matrix, and clusters were formed using the flexible beta linkage method ($\alpha = -0.25$) (Ávila et al., 2011). Prior to the analyses, density data were relativized and square-root transformed to reduce the influence of highly abundant taxa and improve the distributional properties of the dataset (Rocha et al., 2012). The resulting dendrogram was interpreted visually, and the number of groups retained was determined using a cut-off corresponding to 25% or less of the remaining information (Quinn & Keough, 2023). The statistical significance of compositional differences among groups and spatial levels was evaluated using the Multi-Response Permutation Procedure (MRPP) (Biondini et al., 1985; McCune & Grace, 2002). In addition to the probability associated with the test, the MRPP agreement statistic (A) was used to describe within-group homogeneity relative to that expected under random grouping. Both the hierarchical clustering and MRPP analyses were conducted in PC-ORD version 7.07.

Additive diversity partitioning was used to quantify the contribution of within- and between-unit diversity to total zooplankton diversity. In this framework, alpha diversity (α) represents diversity occurring within sampling units, whereas beta diversity (β) represents the contribution of differences among units at successive levels of the spatial hierarchy. The hierarchical structure included three spatial scales. The finest scale consisted of habitat samples within each site ($n = 3$), the intermediate scale corresponded to each site ($n = 18$, for the Barro Branco stream, and $n = 27$, for the Caiana stream), whereas the broadest scale represented differences among the upper and lower river sections ($n = 36$, for the Barro Branco stream, and $n = 54$, for the Caiana stream). Accordingly, total diversity (γ) was decomposed into within-habitat diversity (α) and the successive beta components associated with differences among habitats (β_1), among sites within river reaches (β_2), and among river reaches (β_3). The contribution of each component was

expressed as a percentage of total diversity. To investigate seasonal variation in the hierarchical structure of diversity, additive partitioning was performed independently for the wet and dry seasons and for each stream.

Additive diversity partitioning was conducted in R using the *adipart* function implemented in the **Vegan** package (Oksanen et al., n.d.; R Core Team, 2017). Significance was evaluated using individual-based randomizations under a Type I null model (Crist et al., 2003). For each hierarchical level, 10,000 randomized communities were generated to produce null distributions for the corresponding α - and β -diversity components. The observed diversity components were then compared with their respective null distributions. The probabilities generated by the randomization procedure were interpreted as measures of departure from random expectations. Values below 0.05 indicated that the observed component was significantly greater than expected under the null model, whereas values above 0.95 indicated that the observed value was significantly lower than expected (Gering et al., 2003; Veech et al., 2003).

3 Results

3.1 Environmental variables and nutrient concentration

Water velocity ranged from 0.022 to 0.293 m/s, with the highest values recorded at site 5. Width varied from 1.2 to 7.2 m, while depth ranged from 13.7 to 44.2 cm. Waters were, on average, well oxygenated (from 5.4 to 8.1 mg/L), and cool, with average temperatures from 25.2 to 27.8 °C. Conductivity was high, between 129.0 and 163.3 S/cm, and pH, neutral to slightly acidic (from 5.5 to 7.0). Turbidity was generally low, although higher values were observed at sites 5 and 6, reaching 17.5 and 29.6 NTU, respectively. Elevation ranged from 41.5 to 112.8 m a.s.l., while the slope reached 50° for some sites. Mud and sand were the main substrate components across all sampled sites, with average cover ranging from 30 to 100% and from 0 to 70%, respectively. Habitat structure varied along the stream course. The reference site (site 1) was characterized by high contributions of overhanging vegetation (90%) and exclusively muddy substrates. Site 2 exhibited similar characteristics, whereas site 3 showed a more reduced habitat complexity. Sites 4–6 displayed greater variation in habitat composition. Overall, the habitat structures contributing most were overhanging vegetation, debris, and root masses, although these contributions varied among sites. Aquatic macrophytes and littoral grass were particularly abundant at site 4, contributing on average 35.8% and 39.2%, respectively, whereas root masses represented the dominant habitat structure at site 5 (76.7%). Sand cover increased downstream, reaching 70% at site 6, where overhanging vegetation was markedly reduced (Table 1).

```
1 #TABELA DE HABITAT----
2
3 m_hab_part <- read.csv("res/m_hab_part.csv",
4                       sep = ";", dec = ".",
```

```

5         row.names = 1,
6         header = TRUE,
7         na.strings = NA)
8 t_grps <- read.csv("res/t_grps.csv",
9                 sep = ";", dec = ".",
10                row.names = 1,
11                header = TRUE,
12                na.strings = NA)
13
14 library(dplyr)
15 library(tidyr)
16 m_trab <- m_hab_part %>%
17   rename_with(~ gsub("_", ".", .))
18 m <- m_trab %>%
19   group_by(PontoN = t_grps$PontoN) %>%
20   summarise(across(where(is.numeric),
21                     list(mean = mean, min = min, max = max)),
22             .groups = 'drop') %>%
23   pivot_longer(
24     cols = -c(PontoN),
25     names_to = c("Variable", ".value"),
26     names_sep = "_"
27   )
28
29 m <- as.data.frame(m)
30 m_wide <- m %>%
31   mutate(stat_string = ifelse(Variable == c("m.Vel.m.s"),
32                                paste0(round(mean, 3), " (" , round(min, 3), "-", round(max, 3)
33                                , paste0(round(mean, 1), " (" , round(min, 1), "-", round(max, 1)
34   unite("Location", PontoN, sep = "_") %>%
35   select(Variable, Location, stat_string) %>%
36   pivot_wider(names_from = Location, values_from = stat_string)
37
38 m_wide
39 m_wide <- as.data.frame(m_wide)
40 cols <- colnames(m_wide)
41 col_order <- c(
42   "Variable",
43   cols[cols != "Variable"][
44     order(as.numeric(gsub("Ponto", "", cols[cols != "Variable"])))]
45 ]
46 )
47 m_wide <- m_wide[, c(col_order)] #sequência das colunas

```

```

48 row_order <- c(
49   "w.Cond.uS.cm",
50   "w.Turb.ntu",
51   "w.pH",
52   "m.Elev.m",
53   "m.Vel.m.s",
54   "m.Slope",
55   "m.Depth.avg.cm",
56   "m.Depth.mar.cm",
57   "s.Mud",
58   "s.Sand",
59   "s.Gravel.f",
60   "s.Rocks",
61   "h.Macrophytes",
62   "h.Algae",
63   "h.Subm.veg",
64   "h.Overh.veg",
65   "h.Grass",
66   "h.Leaf.l",
67   "h.Root.masses",
68   "h.Debris"
69 )
70 m_wide <- m_wide[match(row_order, m_wide$Variable), ]
71 m_wide
72
73 # Salvado m_wide
74 write.table(m_wide, "res/m_wide_hab.txt",
75             sep = ";", dec = ".", #"\\t",
76             row.names = TRUE,
77             quote = TRUE,
78             append = FALSE)
79 m_wide_hab <- read.table("res/m_wide_hab.txt",
80                          sep = ";", dec = ".",
81                          row.names = 1,
82                          header = TRUE,
83                          na.strings = NA)
84
85 # Exportando dados para Excel
86 library(openxlsx)
87 #write.xlsx(m_wide, file = "res/tabela de habitat.xlsx", rowNames = FALSE)
88 wb <- loadWorkbook("res/tabela de habitat.xlsx")
89 writeData(wb, sheet = "Sheet 1", x = m_wide)
90 saveWorkbook(wb, "res/tabela de habitat.xlsx", overwrite = TRUE)

```

```

91
92 # Tabela final ajustada no Excel
93 library(readxl)
94 hab_gttable <- read_excel(
95   path = "res/tabela de habitat.xlsx",
96   sheet = "tabela_final",
97   range = "A1:P21") #ajustar para o tamanho da tabela final
98 hab_gttable
99 library(gt)
100 hab_gttable <- gt(hab_gttable)
101 hab_gttable <- sub_missing(hab_gttable,
102   columns = everything(), missing_text = "")
103 #hab_gttable <- fmt_number(hab_gttable, columns = "F.D.(%)", decimals = 1)
104 hab_gttable
105 gtsave(hab_gttable, "res/gt-hab_gttable_xlsx.html")
106 saveRDS(hab_gttable, "res/gt-hab_gttable_xlsx.rds")

1 # Escolher sumário de uma variavel
2 m
3 var <- "w.Temp.C"
4 m[m$Variable == var, "mean"] #cada valor de var
5 summary(m[m$Variable == var, "mean"]) #sumário dos valores de var
6
7 # Escolher sumário de um grupo de variáveis do df m
8 vars <- unique(grep("^h\\.\\.", m$Variable, value = TRUE))
9 summaries <- list() #cria uma lista vazia para guardar os sumários
10 # Loop para cada variável do grupo e guarda em summaries
11 for (var in vars) {
12   summaries[[var]] <- summary(m[m$Variable == var, "mean"])
13 }
14 #var is a temporary variable used in the for loop to iterate through
15 #each variable name that starts with "h."
16
17 summaries
18 summary_table <- do.call(rbind, lapply(summaries, as.data.frame.list))
19 round(summary_table, 2)
20 #sink(file = "summary_h.txt", split = TRUE)
21 round(summary_table[order(summary_table$Mean, decreasing = FALSE), ], 2)
22 #sink()
23
24 # Tabela limpa
25 #summary_table <- cbind(Variable = rownames(summary_table), summary_table)
26 #rownames(summary_table) <- NULL

```

```

27 #colnames(summary_table) <- c("Variable", "Min", "Q1", "Median", "Mean", "Q3", "Max")
28 #summary_table

```

Principal Component Analysis (PCA) was used to summarize environmental variation among sampling sites and to identify variables contributing to their differences (Figure 2 and Figure 3). The first two principal components explained 37% of the total variance, with the first axis accounting for 21.1%. PCA confirmed the expected quality of sites, separating site types (Figure 2), where site types I and II were associated with higher elevation, the presence of algae and leaf litter, and where dominated by muddy substrate. Sites type III, tended to have higher water velocity and the presence of gravel and sand as substrate. The presence of root masses and greater inclination on the margins were also important features of these sites. Sites type IV showed a wide range of environmental characteristics, certainly due to their greater number. Nonetheless, these sites may be associated with greater depths, the presence of rocks in the substrate, and higher pH and turbidity. Comparison of the two study streams showed that, the Barro Branco stream tended to have a greater presence of algae, macrophytes, leaf litter and submerged vegetation as major habitat structures, a muddy substrate and an overall higher elevation. The Caiana stream was characterized by sandy substrate, with occasionally gravel and rocks, higher depths, and higher pH, turbidity, and water velocity. Dry and rainy seasons did not show clear segregation in the PCA.

```

1  #PCA fviz package----
2  #browseURL("https://www.sthda.com/english/articles/31-principal-component-methods-in-r-pract
3
4  #ORGANIZANDO DADOS----
5
6  dev.off()
7  rm(list=ls(all=TRUE))
8  cat("\014")
9
10 t_grps <- read.csv("res/t_grps.csv",
11                   sep = ";", dec = ".",
12                   row.names = 1,
13                   header = TRUE,
14                   na.strings = NA)
15 m_hab_trns <- read.csv("res/m_hab_trns.csv",
16                      sep = ";", dec = ".",
17                      row.names = 1,
18                      header = TRUE,
19                      na.strings = NA)
20
21 #PCA----
22
23 library("FactoMineR")

```

```

24 library("factoextra")
25
26 pca <- PCA(m_hab_trns, scale.unit = TRUE, ncp = 5, graph = TRUE)
27 print(pca)
28
29 eig.val <- get_eigenvalue(pca)
30 eig.val
31
32 #sink(file = "res/cor_matrix.txt", append = FALSE)
33 pca$var$cor
34 #sink()
35
36 fviz_eig(pca, addlabels = TRUE, ylim = c(0,30))
37
38 var <- get_pca_var(pca)
39 var
40
41 fviz_pca_var(pca, col.var = "black")
42
43 library("corrplot")
44 corrplot(var$cos2, is.corr=FALSE)
45
46 fviz_cos2(pca, choice = "var", axes = 1:2)
47
48 fviz_pca_var(pca, col.var = "cos2",
49               gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
50               repel = TRUE # Avoid text overlapping
51 )
52
53 fviz_pca_var(pca, alpha.var = "cos2")
54
55 corrplot(var$contrib, is.corr=FALSE)
56
57 fviz_contrib(pca, choice = "var", axes = 1, top = 10)
58 fviz_contrib(pca, choice = "var", axes = 2, top = 10)
59 fviz_contrib(pca, choice = "var", axes = 1:2, top = 10)
60
61 fviz_pca_var(pca, col.var = "contrib",
62               gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07")
63 )
64 fviz_pca_var(pca, alpha.var = "contrib")
65
66 ##GROUPING BY KMEANS----

```

```

67
68 set.seed(123)
69 res.km <- kmeans(var$coord, centers = 3, nstart = 25)
70 grp <- as.factor(res.km$cluster)
71 # Color variables by groups
72 fviz_pca_var(pca, col.var = grp,
73               palette = c("#0073C2FF", "#EFC000FF", "#868686FF"),
74               legend.title = "Cluster")
75
76 ind <- get_pca_ind(pca)
77 ind
78 ind$contrib
79
80 ##BIPLOTS----
81
82 fviz_pca_ind(pca)
83
84 fviz_pca_ind(pca, col.ind = "cos2",
85               gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
86               repel = TRUE #avoid text overlapping (slow if many points)
87 )
88
89 fviz_pca_ind(pca, pointsize = "cos2",
90               pointshape = 21, fill = "#E7B800",
91               repel = TRUE #avoid text overlapping (slow if many points)
92 )
93
94 fviz_pca_ind(pca, col.ind = "cos2", pointsize = "cos2",
95               gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
96               repel = TRUE #avoid text overlapping (slow if many points)
97 )
98
99 fviz_cos2(pca, choice = "ind")
100
101 fviz_contrib(pca, choice = "ind", axes = 1:2)
102
103 # Create a random continuous variable of length 23,
104 # Same length as the number of active individuals in the PCA
105 t_grps <- t_grps[rownames(pca$ind$coord), ]
106 colnames(t_grps)
107 set.seed(123)
108 # Escolhendo variável categorica
109 my.cont.var <- t_grps$Rio

```

```

110 my.cont.var
111 # Color individuals by the continuous variable
112 fviz_pca_ind(
113   pca,
114   habillage = my.cont.var,
115   palette = c("blue", "yellow", "red", "pink"),
116   repel = FALSE
117 )
118
119 fviz_pca_ind(pca,
120   geom.ind = "point", #show points only (nbut not "text")
121   col.ind = as.factor(my.cont.var), #color by groups
122   palette = "grey",
123   addEllipses = TRUE, #concentration ellipses
124   legend.title = "Groups"
125 )
126
127 length(unique(my.cont.var))
128 fviz_pca_ind(pca,
129   geom.ind = "point",
130   col.ind = my.cont.var,
131   palette = "grey",
132   addEllipses = TRUE, ellipse.type = "confidence",
133   mean.point = TRUE,
134   legend.title = "Groups")
135
136 fviz_pca_biplot(pca, repel = TRUE,
137   col.var = "#2E9FDF", #variables color
138   col.ind = "#696969" #individuals color
139 )
140
141 fviz_pca_biplot(pca,
142   col.ind = my.cont.var, palette = "jco",
143   addEllipses = TRUE, ellipse.type = "t", #t-distribution confidence ellipses"
144   ellipse.level = 0.95,
145   shape.ind = my.cont.var,
146   label = "var",
147   col.var = "black", repel = TRUE,
148   legend.title = "Species") +
149   scale_color_manual(values = c(
150     "#1B9E77",
151     "#D95F02",
152     "#7570B3",

```



```

153     "#E7298A"
154   )) +
155   scale_shape_manual(values = c(
156     16, # circle
157     17, # triangle
158     15, # square
159     18 # diamond
160   ))
161
162   ### FILTRANDO AS VARIÁVEIS
163   var_coord <- pca$var$coord[,1:2]
164   keep_vars <- apply(abs(var_coord) > 0.5,1,any)
165   vars_to_keep <- rownames(var_coord)[keep_vars]
166   pca_filt <- pca
167   pca_filt$var$coord <- pca$var$coord[vars_to_keep,,drop=FALSE]
168   pca_filt$var$cos2 <- pca$var$cos2[vars_to_keep,,drop=FALSE]
169   pca_filt$var$contrib <- pca$var$contrib[vars_to_keep,,drop=FALSE]
170
171   ### AJUSTE MANUAL DAS COORDENADAS DOS VETORES
172   var_coords <- as.data.frame(pca_filt$var$coord)
173   var_coords$Dim.1 <- var_coords$Dim.1*4.2
174   var_coords$Dim.2 <- var_coords$Dim.2*4.2
175
176   ## Caso queira editar manualmente
177   #fix(var_coords) #nomes das colunas
178   #write.table(var_coords,"coords_var.txt")
179
180   var_coords <- read.table("coords_var.txt",
181                           header=TRUE,
182                           row.names=1)
183
184   ### AJUSTE MANUAL DOS INDIVÍDUOS
185   ind_coords <- as.data.frame(pca_filt$ind$coord)
186
187   ## Caso queira editar manualmente
188   #fix(ind_coords) #nomes das linhas
189   #write.table(ind_coords,"coords_ind.txt")
190
191   ind_coords <- read.table("coords_ind.txt",
192                           header=TRUE,
193                           row.names=1)
194
195   # GRÁFICO FINAL

```

```

196
197 library(factoextra)
198 library(ggplot2)
199
200 t_grps$Ref_site <- as.factor(t_grps$Ref_site)
201
202 #png("fig-hab_PCA_fviz.png", width=11, height=9, units="in", res=400)
203
204 library(ggplot2)
205
206 ggplot() +
207
208   geom_hline(yintercept = 0, linetype = 2) +
209   geom_vline(xintercept = 0, linetype = 2) +
210
211   geom_point(
212     data = ind_coords,
213     aes(Dim.1, Dim.2, shape = t_grps$Ref_site),
214     size = 3,
215     colour = "black"
216   ) +
217
218   geom_segment(
219     data = var_coords,
220     aes(x = 0, y = 0,
221         xend = Dim.1,
222         yend = Dim.2),
223     colour = "red",
224     arrow = arrow(length = unit(0.25, "cm"))
225   ) +
226
227   geom_text(
228     data = transform(
229       var_coords,
230       nudge_y = ifelse(Dim.2 >= 0, 0.25, -0.25)
231     ),
232     aes(
233       x = Dim.1,
234       y = Dim.2 + nudge_y,
235       label = rownames(var_coords)
236     ),
237     colour = "red",
238     size = 4,

```

```

239     nudge_x = 0.19
240   ) +
241
242   scale_shape_manual(values = c(15,16,0,1,2,17)) +
243
244   labs(
245     x = paste0("Dim1 (", round(pca$eig[1,2],1), "%)" ),
246     y = paste0("Dim2 (", round(pca$eig[2,2],1), "%)" ),
247     shape = "Sites"
248   ) +
249
250   coord_equal() +
251   theme_minimal()
252
253   #dev.off()

```

1 *#ORGANIZANDO DADOS----*

2

```

3   dev.off()
4   rm(list=ls(all=TRUE))
5   cat("\014")
6
7   t_grps <- read.csv("res/t_grps.csv",
8                     sep = ";", dec = ".",
9                     row.names = 1,
10                    header = TRUE,
11                    na.strings = NA)
12   m_hab_trns <- read.csv("res/m_hab_trns.csv",
13                        sep = ";", dec = ".",
14                        row.names = 1,
15                        header = TRUE,
16                        na.strings = NA)
17
18   colnames(t_grps)
19   m_trab <- m_hab_trns
20   library(vegan)
21
22   #MRPP----
23
24   #TRABALHANDO COM MULTIPLAS VARIAVEIS-----
25
26   # Run MRPP for multiple variables
27   fatores <- c("Periodo", "Rio", "Ref_site")

```

```

28 set.seed(123)
29 mrpp_results <- lapply(fatores, function(v) {
30   grp <- as.factor(t_grps[[v]])
31   mrpp(
32     dat = m_trab,
33     grouping = grp,
34     distance = "bray"
35   )
36 })
37 names(mrpp_results) <- fatores
38 mrpp_results
39 # Print one result
40 mrpp_results$Ref_site
41 # Loop through summaries
42 sink(file = "res/mrpp.txt", split = TRUE)#, append = TRUE)
43 for (v in names(mrpp_results)) {
44   cat("\n=====\\n")
45   cat("MRPP for:", v, "\\n")
46   print(mrpp_results[[v]])
47 }
48 sink()
49 # Extract key statistics into a table
50 mrpp_summary <- data.frame(
51   Variable = fatores,
52   Delta = sapply(mrpp_results, function(x) x$delta),
53   Expected_Delta = sapply(mrpp_results, function(x) x$E.delta),
54   A = sapply(mrpp_results, function(x) x$A),
55   P_value = sapply(mrpp_results, function(x) x$Pvalue)
56 )
57 sink("res/mrpp.txt", append = TRUE)
58 mrpp_summary
59 sink()
60
61 ##TRABALHANDO COM UMA UNICA VARIABEL-----
62
63 # Run MRPP for a single variable
64 grp <- as.factor(t_grps$Ref_site) #UA, mes, estacao, area, ambiente
65 set.seed(123)
66 mrpp <- mrpp(dat = m_trab, grouping = grp, distance = "bray")
67 mrpp
68 summary(mrpp)
69
70 # Teste de hipótese para uma unica variavel

```

```

71 str(mrpp)
72 p_value <- mrpp$Pvalue
73 print(sprintf("p-value: %.10f", p_value))
74 # Conditional statement to check the p-value and print the appropriate message
75 if (p_value < 0.05) {
76   print(sprintf("Existe diferença significativa porque o valor de p é %.10f, sendo menor que
77 } else {
78   print(sprintf("Não existe diferença significativa porque o valor de p é %.10f, sendo maior
79 }
80
81 ##TESTES PAREADOS----
82
83 # Todos os pares de níveis do fator
84 pairs <- combn(levels(as.factor(grp)), 2, simplify = FALSE)
85 pairs
86 # Aplicar mrpp para cada par
87 res_list <- lapply(pairs, function(p) {
88   sel <- grp %in% p
89   mrpp(dat = m_trab[sel, ], grouping = droplevels(grp[sel]), distance = "bray")
90 })
91 # Dar nomes aos resultados
92 names(res_list) <- sapply(pairs, function(p) paste(p, collapse = "_vs_"))
93 # visualizar
94 res_list
95 # Resumo
96 summary_tab <- data.frame(
97   Comparison = names(res_list),
98   A = sapply(res_list, function(x) x$A),
99   Delta = sapply(res_list, function(x) x$delta),
100   Pvalue = sapply(res_list, function(x) x$Pvalue)
101 )
102 sink("res/mrpp.txt", append = TRUE)
103 summary_tab
104 sink()
105
106 ###MATRIX-STYLE TABLE----
107 library(dplyr)
108 library(tidyr)
109 library(knitr)
110 library(kableExtra)
111 # Extract unique group names from the comparisons
112 groups1 <- sort(unique(unlist(strsplit(summary_tab$Comparison, "_vs_"))))
113 # Initialize empty matrix

```

```

114 pval_matrix <- matrix(NA, nrow = length(groups1), ncol = length(groups1),
115                       dimnames = list(groups1, groups1))
116 # Fill matrix with Pvalues
117 for (i in 1:nrow(summary_tab)) {
118   comp <- unlist(strsplit(summary_tab$Comparison[i], "_vs_"))
119   g1 <- comp[1]
120   g2 <- comp[2]
121   p <- summary_tab$Pvalue[i]
122
123   pval_matrix[g1, g2] <- p
124   pval_matrix[g2, g1] <- p #make symmetric
125 }
126
127 # Format p-values: red if <0.05
128 pval_colored <- ifelse(is.na(pval_matrix), "",
129                       ifelse(pval_matrix < 0.05,
130                             paste0("<span style='color:red;'>", sprintf("%.3f", pval_matrix[i, j])),
131                             sprintf("%.3f", pval_matrix[i, j])))
132
133 # Convert to data.frame for kable
134 pval_df <- as.data.frame(pval_colored)
135 pval_df <- tibble::rownames_to_column(pval_df, "Group")
136 # Render HTML table
137 matrixstyle <- pval_df %>%
138   kable("html", escape = FALSE) %>%
139   kable_styling(full_width = FALSE)
140 matrixstyle
141 save_kable(matrixstyle, "res/kb-mrpp_matrixstyle.html")
142
143 ##GRÁFICO DA MRPP----
144
145 png("res/fig-mrpp.png", width = 15, height = 10, units = "in", res = 300)
146 def.par <- par(no.readonly = TRUE)
147 layout(matrix(1:2,nr=1))
148 plot(ord <- metaMDS(m_trab, distance="bray"), type="text", display="sites")
149 with(t_grps, ordihull(ord, grp))
150 with(mrpp, {
151   fig.dist <- hist(boot.deltas, xlim=range(c(delta,boot.deltas)),
152                   main="Test of Differences Among Groups")
153   abline(v=delta);
154   text(delta, 2*mean(fig.dist$counts), adj = -0.5,
155         expression(bold(delta)), cex=1.5 ) }
156 )

```

```

157 par(def.par)
158 dev.off()
159
160 ###AVALIANDO AS DISTÂNCIAS MÉDIAS----
161 md <- with(t_grps, meandist(vegdist(m_trab), grp))
162 md
163 summary(md)
164 par(mfrow=c(1,2))
165 plot(md)
166 plot(md, kind="histogram")
167 par(mfrow=c(1,1))
168
169 #PerMANOVA----
170
171 #browseURL("https://riffomonas.org/code_club/2021-03-18-vegan")
172 #browseURL("https://riffomonas.org/code_club/2021-03-22-reproducibility")
173
174 library(tidyverse)
175 library(ggtext)
176 library(ggplot2)
177 library(vegan)
178
179 adonis2(m_trab ~ Perodo, data = t_grps, method = "bray")
180 set.seed(123)
181 perm <- 999
182
183 all_test <- adonis2(m_trab ~ Perodo, data = t_grps, method = "bray", permutations = perm)
184 all_test
185
186 adonis2(m_trab ~ Perodo*Rio, data = t_grps)
187 adonis2(m_trab ~ Perodo*Rio, data = t_grps, by = "terms") #ou by=NULL remove as interações
188
189 all_p <- all_test$`Pr(>F)`[1]
190 all_p
191
192 count(t_grps, Perodo)
193 colnames(t_grps)
194
195 # Pairwise comparisons
196 #browseURL("https://doi.org/10.1007/s10641-018-0751-1")
197 #Lanés, L.E.K., Reichard, M., de Moura, R.G. et al. 2018.
198 #Environmental predictors for annual fish assemblages in subtropical
199 #grasslands of South America: the role of landscape and habitat

```

```

200 #characteristics. Environ Biol Fish 101, 963-977.
201
202 library(dplyr)
203 #library(combnat) # or base R combn()
204
205 set.seed(123)
206
207 # Get unique levels of UA
208 groups <- unique(t_grps$Ref_site)
209 groups
210 # All pairwise combinations
211 pairs <- combn(groups, 2, simplify = FALSE)
212 pairs
213 # Store results
214 pairwise_results <- lapply(pairs, function(g) {
215   # Subset data
216   sub_data <- t_grps %>% filter(Ref_site %in% g)
217   sub_dist <- m_trab[rownames(sub_data), ]
218
219   # Run adonis2
220   test <- adonis2(sub_dist ~ Ref_site, data = sub_data,
221                   method = "bray", permutations = perm)
222
223   tibble(
224     group1 = g[1],
225     group2 = g[2],
226     F_value = test$F[1],
227     R2 = test$R2[1],
228     p_value = test$`Pr(>F)`[1]
229   )
230 })
231 # Bind all results
232 pairwise_results <- bind_rows(pairwise_results)
233 pairwise_results
234 pairwise_p <- numeric()
235 pairwise_p <- pairwise_results$p_value
236 pairwise_p
237 adj_p_values <- p.adjust(pairwise_p, method="bonferroni") #c("holm", "hochberg", "hommel", "
238 adj_p_values
239 pairwise_results$adj_p_values <- adj_p_values
240 pairwise_results
241 # Get unique group names
242 groups2 <- sort(unique(c(pairwise_results$group1, pairwise_results$group2)))

```



```

243 # Initialize empty matrix
244 pval_matrix <- matrix(NA, nrow = length(groups2), ncol = length(groups2),
245                       dimnames = list(groups2, groups2))
246 # Fill in p-values
247 for (i in 1:nrow(pairwise_results)) {
248   g1 <- pairwise_results$group1[i]
249   g2 <- pairwise_results$group2[i]
250   p <- pairwise_results$p_value[i]
251
252   pval_matrix[g1, g2] <- p
253   pval_matrix[g2, g1] <- p #make symmetric
254 }
255 pval_matrix
256
257 library(knitr)
258 library(kableExtra)
259
260 # Make a copy of the matrix
261 pval_colored <- pval_matrix
262 # Replace numeric values with formatted HTML strings
263 pval_colored[] <- ifelse(is.na(pval_matrix), "",
264                          ifelse(pval_matrix < 0.05,
265                                paste0("<span style='color:red;'>", sprintf("%.3f", pval_matr
266                                sprintf("%.3f", pval_matrix)))
267 # Convert to data frame with rownames
268 pval_df <- as.data.frame(pval_colored)
269 pval_df <- tibble::rownames_to_column(pval_df, "Group")
270 # Render table
271 pval_df %>%
272   kable("html", escape = FALSE) %>%
273   kable_styling(full_width = FALSE)
274 save_kable(matrixstyle, "res/kb-perma_matrixstyle.html")
275
276 ## AVALIANDO INTERAÇÕES-----
277
278 adonis2(m_trab ~ Periodo, data = t_grps, method = "bray")
279 adonis2(m_trab ~ Periodo*Rio, data = t_grps,
280        method = "bray") #+ no lugar de * tira os residuos
281 adonis2(m_trab ~ Periodo*Rio, data = t_grps,
282        method = "bray", by = "terms", strata = NULL)

```

Among the averaged nutrient concentrations (Table 2), ammonia ranged from 49.2 to 243.7 $\mu\text{g L}^{-1}$. The Caiana stream tended to have greater values, overall no clear trend can be

observed when comparing both streams. Nitrite and nitrate, on the other hand, show greater concentrations at the Caiana stream, the latter varying from 166.4 to 2003.0 $\mu\text{g L}^{-1}$, compared with the maximum of 18.8 $\mu\text{g L}^{-1}$ at the Barro Branco stream. Nitrite was too low to be detectable at the Barro Branco stream. The same pattern was observed for orthophosphate concentrations, being barely detectable at Barro Branco stream, and varying from 4.0 to 13.0 $\mu\text{g L}^{-1}$ at the Caiana stream. Total phosphorus, however, tended to have higher concentrations at Barro Branco stream, reaching 50.1 $\mu\text{g L}^{-1}$ at Site 3, but with highly variable values (from non detectable to 38.7 $\mu\text{g L}^{-1}$, as the second highest value).

```

1  #TABELA DE NUTRIENTES----
2
3  ##ORGANIZANDO DADOS----
4
5  dev.off()
6  rm(list=ls(all=TRUE))
7  cat("\014")
8
9  t_nutri <- read.csv("res/t_nutri.csv",
10                    sep = ";", dec = ".",
11                    row.names = 1,
12                    header = TRUE,
13                    na.strings = NA)
14
15  t_grps <- read.csv("res/t_grps.csv",
16                   sep = ";", dec = ".",
17                   row.names = 1,
18                   header = TRUE,
19                   na.strings = NA)
20
21  nutri <- t_nutri
22
23  library(dplyr)
24  library(tidyr)
25  t_grps <- t_grps[seq_len(nrow(nutri)), , drop = FALSE]
26  m_trab <- nutri %>%
27    rename_with(~ gsub("_", ".", .))
28  m <- m_trab %>%
29    group_by(PontoN = t_grps$PontoN) %>%
30    summarise(across(where(is.numeric),
31                     list(mean = mean, min = min, max = max)),
32              .groups = 'drop') %>%
33    pivot_longer(
34      cols = -c(PontoN),

```

```

35     names_to = c("Variable", ".value"),
36     names_sep = "_"
37 )
38
39 m <- as.data.frame(m)
40 colnames(m_trab)
41 m_wide <- m %>%
42   mutate(stat_string = ifelse(Variable == c("Ammonia.NH3.ug.L"),
43                               paste0(round(mean, 1), " (", round(min, 1), "-", round(max, 1)
44                               paste0(round(mean, 1), " (", round(min, 1), "-", round(max, 1)
45   unite("Location", PontoN, sep = "_") %>%
46   select(Variable, Location, stat_string) %>%
47   pivot_wider(names_from = Location, values_from = stat_string)
48
49 m_wide
50 m_wide <- as.data.frame(m_wide)
51 cols <- colnames(m_wide)
52 col_order <- c(
53   "Variable",
54   cols[cols != "Variable"][
55     order(as.numeric(gsub("Ponto", "", cols[cols != "Variable"])))
56   ]
57 )
58 m_wide <- m_wide[, c(col_order)] #sequência das colunas
59 m_wide
60
61 # Salvado m_wide
62 write.table(m_wide, "res/m_wide_nutri.txt",
63             sep = ";", dec = ".", #"\\t",
64             row.names = TRUE,
65             quote = TRUE,
66             append = FALSE)
67 m_wide_nutri <- read.table("res/m_wide_nutri.txt",
68                            sep = ";", dec = ".",
69                            row.names = 1,
70                            header = TRUE,
71                            na.strings = NA)
72
73 # Exportando dados para Excel
74 library(openxlsx)
75 #write.xlsx(m_wide, file = "res/tabela de nutrientes.xlsx", rowNames = FALSE)
76 wb <- loadWorkbook("res/tabela de nutrientes.xlsx")
77 writeData(wb, sheet = "Sheet 1", x = m_wide_nutri)

```

```

78 saveWorkbook(wb, "res/tabela de nutrientes.xlsx", overwrite = TRUE)
79
80 # Tabela final ajustada no Excel
81 library(readxl)
82 nutri_gtttable <- read_excel(
83   path = "res/tabela de nutrientes.xlsx",
84   sheet = "tabela_final",
85   range = "A1:P6") #ajustar para o tamanho da tabela final
86 nutri_gtttable
87 library(gt)
88 nutri_gtttable <- gt(nutri_gtttable)
89 nutri_gtttable <- sub_missing(nutri_gtttable,
90                               columns = everything(), missing_text = "")
91 nutri_gtttable
92 gtsave(nutri_gtttable, "res/gt-nutri_gtttable_xlsx.html")
93 saveRDS(nutri_gtttable, "res/gt-nutri_gtttable_xlsx.rds")

```

3.2 Community structure

```

1  #ZOOEBIO_part - CURVA DE ACUMULAÇÃO----
2  #####....----
3
4  ##ORGANIZANDO DADOS----
5
6  dev.off() #apaga os graficos, se houver algum
7  rm(list=ls(all=TRUE)) #limpa a memória
8  cat("\014") #limpa o console
9
10 m_cont <- read.csv("res/m_cont.csv",
11                   sep = ";", dec = ".",
12                   row.names = 1,
13                   header = TRUE,
14                   na.strings = NA)
15 m_trab <- m_cont
16
17 library(tidyverse)
18
19 m_trab_r <- mutate(m_trab, across(everything(), ceiling)) #arredonda a matriz original para
20
21 colSums(t(m_trab_r))
22 colnames(t(m_trab_r))
23

```

```

24 ca_abund <- colSums(m_trab_r[grepl("-", rownames(m_trab_r)), ])
25 bb_abund <- colSums(m_trab_r[grepl("-", rownames(m_trab_r)), ])
26
27 m_rios <- rbind("Caiana" = ca_abund, "Barro Branco" = bb_abund)
28 m_rios
29
30 rarefa <- t(m_rios)
31
32 library(iNEXT)
33 out <- iNEXT(rarefa, q = 0,
34             datatype = "abundance",
35             size = NULL,
36             endpoint = 15000, #define o comprimento de eixo x
37             knots = 40,
38             se = TRUE,
39             conf = 0.95,
40             nboot = 50)
41
42 library(ggplot2)
43 grafico3 <- ggiNEXT(out, type = 1, facet.var="None") + #ver como fica com facet.var="Assembl
44   theme_bw() +
45   labs(fill = "Áreas") +
46   xlab("Número de indivíduos") +
47   ylab("Riqueza de espécies") +
48   scale_linetype_discrete(
49     labels = c("Interpolation", "Extrapolation")) +
50   theme(legend.title = element_blank(),
51         legend.position = "bottom",
52         legend.text = element_text(size = 14),
53         axis.title = element_text(size = 14),
54         axis.text = element_text(size = 12)
55   )
56 grafico3
57 ggsave(grafico3, width = 8, height = 6, dpi = 300, filename = "res/fig-rare1.png")

1 #ZOOBIO_part - ESTR. DA COMUNIDADE----
2 #####....----
3
4 ##ORGANIZANDO DADOS----
5
6 dev.off() #apaga os graficos, se houver algum
7 rm(list=ls(all=TRUE)) #limpa a memória
8 cat("\014") #limpa o console

```

```

9
10 t_grps <- read.csv("res/t_grps.csv",
11                   sep = ";", dec = ".",
12                   row.names = 1,
13                   header = TRUE,
14                   na.strings = NA)
15 m_dens <- read.csv("res/m_dens.csv",
16                   sep = ";", dec = ".",
17                   row.names = 1,
18                   header = TRUE,
19                   na.strings = NA)
20 m_hab_part <- read.csv("res/m_hab_part.csv",
21                       sep = ";", dec = ".",
22                       row.names = 1,
23                       header = TRUE,
24                       na.strings = NA)
25 m_cont <- read.csv("res/m_cont.csv",
26                   sep = ";", dec = ".",
27                   row.names = 1,
28                   header = TRUE,
29                   na.strings = NA)
30
31 library(tibble)
32 library(tidyverse)
33 library(forcats)
34 library(openxlsx)
35 library(Rcpp)
36
37 m_dens
38 str(m_dens)
39 m_trab <- m_dens
40
41 m_trab <- (m_dens) #SITES
42 m_trab <- t(m_dens) #SPP
43
44 ##TAMANHO DA MATRIZ----
45
46 range(m_trab) #menor e maior valores
47 length(m_trab) #no. de colunas
48 ncol(m_trab) #no. de N colunas
49 nrow(m_trab) #no. de M linhas
50 sum(lengths(m_trab)) #soma os nos. de colunas
51 length(as.matrix(m_trab)) #tamanho da matriz m x n

```

```

52 sum(m_trab == 0) #número de observações igual a zero
53 sum(m_trab > 0) #número de observações maiores que zero
54 #calculando a proporção de zeros na matriz
55 zeros <- (sum(m_trab == 0)/length(as.matrix(m_trab)))*100
56 zeros
57
58 ##ESTRUTURA DA COMUNIDADE-----
59
60 Sum <- rowSums(m_trab)
61 #ou
62 Sum <- apply(m_trab,1,sum)
63 Sum
64 # Abundância relativa (%)
65 RA <- (Sum / sum(Sum)) * 100 # percentage
66 # Media
67 Mean <- rowMeans(m_trab)
68 Mean
69 # Ou
70 Mean <- apply(m_trab,1,mean)
71 Mean
72 # Desvio padrão
73 DP <- apply(m_trab,1,sd)
74 DP
75 # Máximo
76 Max <- apply(m_trab,1,max)
77 Max
78 # Mínimo
79 Min <- apply(m_trab,1,min)
80 Min
81 # Mínimo não-zero
82 MinZ <- apply(m_trab, 1, function(row) {
83   non_zero_values <- row[row > 0] # Filter out zero values
84   if (length(non_zero_values) == 0) {
85     return(0) # If all values are zero, return 0
86   } else {
87     return(min(non_zero_values)) # Return the minimum of non-zero values
88   }
89 })
90 MinZ
91
92 m_pa <- m_trab
93 m_pa[m_pa != 0] <- 1
94 rowSums(m_pa)

```

```

95 library(vegan)
96 bin <- decostand(m_trab,"pa")
97 bin[1:10, 1:10]
98 S <- apply(bin,1,sum)
99 S
100 #OU
101 Riqueza <- specnumber(m_trab)
102 Riqueza
103 Riqueza_total <- specnumber(colSums(m_trab))
104 Riqueza_total
105
106 FO <- rowSums(m_trab > 0) / ncol(m_trab) * 100
107 FO
108
109 H <- diversity(m_trab, index = "shannon")
110 H
111 D <- diversity(m_trab, "simpson")
112 D
113 D[is.na(D)] <- 0 #substitui NA ou NaN por 0
114 D
115 E <- H/log(specnumber(m_trab))
116 E
117 E[is.na(E)] <- 0 #substitui NA ou NaN por 0
118 E
119
120 ###DESCRITORES----
121 Descritores1 <- cbind(Sum, RA, Mean, DP, Max, Min, MinZ, FO, S, E, H, D)
122 Descritores1 <- as.data.frame(Descritores1)
123 Descritores1
124 Descritores1 <- Descritores1[order(-Descritores1$Mean), ] #ordena do maior para o menor "-"
125 #Descritores1 <- Descritores1 %>% rownames_to_column(var="Espécies") #da nome a primeira col.
126 SomaTotalD <- apply(Descritores1,2,sum)
127 SomaTotalD
128 MediaTotalD <- apply(Descritores1,2,mean)
129 MediaTotalD
130 DPTotalD <- apply(Descritores1,2,sd)
131 DPTotalD
132 Descritores2 <- cbind(SomaTotalD, MediaTotalD, DPTotalD)
133 Descritores2 <- as.data.frame(Descritores2)
134 Descritores2 <- t(Descritores2)
135 Descritores2
136 DescritoresFinal <- rbind(Descritores1, Descritores2)
137 DescritoresFinal

```



```

138 DescritoresFinal <- round (DescritoresFinal, 2)
139 DescritoresFinal
140
141 # Salvando DescritoresFinal
142 write.table(data.frame("SPP"=rownames(DescritoresFinal),
143                       DescritoresFinal),
144             "res/DescritoresSPP.txt",
145             row.names=FALSE,
146             sep="\t")
147
148 ###TABELA FINAL DESCRITORES-----
149 library(gt)
150 df <- DescritoresFinal
151 ncol(df); nrow(df) #no. de N colunas x M linhas
152 df <- cbind(SPP = rownames(df), df)
153 gt <- gt(df, rowname_col = "SPP",
154          caption = "Descritores da diversidade por espécie (colunas). Sum, soma; RA, abundância re
155 gt
156 gtsave(gt, "res/DescritoresSPP.html")
157 gtsave(gt, "res/DescritoresSPP.png")
158
159 ###NORMALIDADE-----
160 library(moments)
161 Assimetria <- apply(m_trab,1,skewness)
162 Assimetria
163 Curtose <- apply(m_trab,1,kurtosis)
164 Curtose
165 Normalidade1 <- cbind(Assimetria, Curtose)
166 Normalidade1 <- as.data.frame(Normalidade1)
167 Normalidade1
168 SomaTotalN <- apply(Normalidade1,2,sum)
169 SomaTotalN
170 MediaTotalN <- apply(Normalidade1,2,mean)
171 MediaTotalN
172 DPTotalN <- apply(Normalidade1,2,sd)
173 DPTotalN
174 Normalidade2<-cbind(SomaTotalN, MediaTotalN, DPTotalN)
175 Normalidade2<-as.data.frame(Normalidade2)
176 Normalidade2 <- t(Normalidade2) #"t" transpoe a matriz
177 Normalidade2
178 NormalidadeFinal <- rbind(Normalidade1, Normalidade2)
179 NormalidadeFinal
180 NormalidadeFinal <- round(NormalidadeFinal, 2)

```

```

181 NormalidadeFinal
182
183 # Salvando NormalidadeFinal
184 write.table(data.frame("SPP"=rownames(NormalidadeFinal),
185                        NormalidadeFinal),
186            "res/NormalidadeSPP.txt",
187            row.names=FALSE,
188            sep="\t")
189
190 ###TABELA FINAL NORMALIDADE-----
191 nf <- NormalidadeFinal
192 ncol(nf); nrow(nf) #no. de N colunas x M linhas
193 nf <- cbind(SPP = rownames(nf), nf)
194 gt <- gt(nf, rowname_col = "SPP", caption = "Descritores da normalidade por SPP (COLUNA)")
195 gt
196 gtsave(gt, "res/NormalidadeSPP.html")
197
198 #DIVERSIDADE EM DENSIDADE MATRIZ NÃO-TRANSPOSTA-----
199
200 m_trab <- (m_dens) #HILL
201
202 # Números de Hill
203 Hill <- renyi(m_trab,
204             scales = c(0:5),
205             hill = TRUE)
206 Hill
207 library(ggplot2)
208 library(tidyr)
209 library(tidyverse)
210 grafico1 <- Hill %>%
211   rownames_to_column() %>%
212   pivot_longer(-rowname) %>%
213   mutate(name = factor(name, name[1:length(Hill)])) %>%
214   ggplot(aes(x = name, y = value, group = rowname,
215             col = rowname)) +
216   geom_point(size = 2) +
217   geom_line(size = 1) +
218   xlab("Parâmetro de ordem de diversidade (q)") +
219   ylab("Diversidade") +
220   labs(col = "Locais") +
221   theme_bw() +
222   theme(text = element_text(size = 10)) #ajustar a fonte caso nao caiba no output html
223 grafico1

```

```

224 ggsave(grafico1, dpi = 300, filename = "res/fig-hill.png")
225
226 # Gráfico de abundância
227 abund <- colSums(m_trab)
228 #abund <- m_trab[1, ] #escolhe a primeira linha para a distribuição de abundância
229 df <- data.frame(sp = colnames(m_trab),
230                 abun = abund)
231 grafico2 <- ggplot(df, aes(fct_reorder(sp, -abun),
232                               abun, group = 1)) +
233   geom_col() +
234   geom_line(col = "red", linetype = "dashed") +
235   geom_point(col = "red") +
236   xlab("Espécies") +
237   ylab("Abundância") +
238   theme_bw() +
239   theme(axis.text.x = element_text(
240     angle = 45,
241     hjust = 1,
242     face = "italic",
243     size = 10),
244     axis.text.y = element_text(size = 10),
245     axis.title.x = element_text(size = 12),
246     axis.title.y = element_text(size = 12))
247 grafico2
248 ggsave(grafico2, dpi = 300, filename = "res/fig-abun.png")
249
250 # Curvas de rarefação
251 m_trab_r <- mutate(as.data.frame(m_trab), across(everything(), ceiling)) #arredonda a matriz
252 colSums(t(m_trab_r))
253 colnames(t(m_trab_r))
254 #rarefa <- t(m_trab_r[c("S-R-CT2", "S-R-CP2", "S-A-TA2", "B-A-MU2"),]) #curva de rarefação para
255 class(m_trab)
256
257 # Curva de rarefação para todos os sítios
258 rarefa <- t(m_trab_r)
259 ## Curva de rarefação para as 8 UA's com maior soma
260 #m_trab_r <- as.data.frame(t(m_trab_r)) #transpõe a matriz
261 #col_sums <- colSums(m_trab_r)
262 #largest_columns <- names(sort(col_sums, decreasing = TRUE)[1:8])
263 #rarefa <- m_trab_r[largest_columns] #curva de rarefação para as 8 UA's com maior soma
264 library(iNEXT)
265 out <- iNEXT(rarefa, q = 0,
266             datatype = "abundance",

```

```

267         size = NULL,
268         endpoint = 2000, #define o comprimento de eixo x
269         knots = 40,
270         se = TRUE,
271         conf = 0.95,
272         nboot = 50)
273 grafico3 <- ggiNEXT(out, type = 1, facet.var="None") +
274   theme_bw() +
275   labs(fill = "Áreas") +
276   xlab("Número de indivíduos") +
277   ylab("Riqueza de espécies") +
278   theme(legend.title=element_blank()) #ver como fica com facet.var="Assemblage"
279 #grafico3
280 #ggsave(grafico3, dpi = 300, filename = "fig-rare1.png")
281
282 # Curvas de acumulação de espécies
283 acumula <- specaccum(m_trab,
284                     method = "random")
285
286 acumula
287 plot(acumula)
288 plot_data <- data.frame("UAs" = c(0, acumula$sites),
289                        "Riqueza" = c(0, acumula$richness),
290                        "lower" = c(0, acumula$richness - acumula$sd),
291                        "upper" = c(0, acumula$richness + acumula$sd))
292
293 gLocais <- ggplot(plot_data, aes(x = UAs, y = Riqueza)) +
294   geom_point(color = "blue", size = 4) +
295   geom_line(color = "blue", lwd = 2) +
296   geom_ribbon(aes(ymin = lower, ymax = upper),
297             linetype=2, alpha=0.3, fill = "yellow") +
298   ylab("Riqueza acumulada") +
299   theme_classic() +
300   theme(text = element_text(size = 16))
301
302 gLocais
303 ggsave(gLocais, dpi = 300, filename = "res/fig-acum.png")
304
305 plot_data <- data.frame("Individuos" = c(0, acumula$individuals),
306                        "Riqueza" = c(0, acumula$richness),
307                        "lower" = c(0, acumula$richness - acumula$sd),
308                        "upper" = c(0, acumula$richness + acumula$sd))
309
310 gInd <- ggplot(plot_data, aes(x = Individuos, y = Riqueza)) +
311   geom_point(color = "blue", size = 4) +
312   geom_line(color = "blue", lwd = 2) +
313   geom_ribbon(aes(ymin = lower, ymax = upper),

```

```

310         linetype=2, alpha=0.3, fill = "yellow") +
311     ylab("Riqueza acumulada") +
312     theme_classic() +
313     theme(text = element_text(size = 16))
314 gInd
315
316 #RIQUEZA RAREFACIONADA----
317
318 # Rarefação baseada na matriz de contagem
319 library(vegan)
320 data <- m_cont #raremax precisa de numeros inteiros
321 S <- specnumber(data) #observed number of species
322 S
323 #raremax <- min(rowSums(data))
324 raremax <- round(mean(rowSums(data)))
325 #raremax <- 1000
326 raremax
327 Srare <- rarefy(data, raremax)
328 Srare
329 plot(S, Srare, xlab = "Observed No. of Species", ylab = "Rarefied No. of Species")
330 abline(0, 1)
331
332 #png("res/fig-rarecurve.png")
333 rarecurve(data, step = 20,
334           sample = raremax, #linhas de rarefação
335           col = "blue", cex = 0.6,
336           xlab = "Sample size", ylab = "Taxa", xlim = c()) #c(0,5000)
337 #dev.off()
338
339 ##TABELA DA COMUNIDADE----
340
341 library(dplyr)
342 library(tidyr)
343 m_trab <- m_dens %>%
344   rename_with(~ gsub("_", ".", .))
345 m <- m_trab %>%
346   group_by(Rio = t_grps$Rio,
347            Periodo = t_grps$Periodo,
348            Ponto = t_grps$PontoN) %>%
349   summarise(across(where(is.numeric),
350                    list(mean = mean, sd = sd)),
351            .groups = 'drop') %>%
352   pivot_longer(

```

```

353     cols = -c(Rio, Período, Ponto),
354     names_to = c("Variable", ".value"),
355     names_sep = "_"
356   )
357   m
358   m <- as.data.frame(m)
359   m_wide <- m %>%
360     mutate(stat_string = ifelse(
361       Variable == "nome_da_variavel", #escolhe uma variável pra ter 3 casas decimais
362       paste0(round(mean, 3), " (", round(sd, 3), ")"),
363       paste0(round(mean, 1), " (", round(sd, 1), ")")
364     )) %>%
365     unite("Location", Rio, Período, Ponto, sep = "_") %>%
366     dplyr::select(Variable, Location, stat_string) %>% #dplyr dá conflito com MASS
367     pivot_wider(names_from = Location, values_from = stat_string)
368
369   m_wide
370   m_wide <- as.data.frame(m_wide)
371   m_wide
372
373   # Salvado m_wide
374   write.table(m_wide, "res/m_wide_spp.txt",
375     sep = ";", dec = ".", #"\\t",
376     row.names = TRUE,
377     quote = TRUE,
378     append = FALSE)
379   m_wide_spp <- read.table("res/m_wide_spp.txt",
380     sep = ";", dec = ".",
381     row.names = 1,
382     header = TRUE,
383     na.strings = NA)
384
385   # Exportando dados para Excel
386   library(openxlsx)
387   #write.xlsx(m_wide_spp, file = "res/tabela da comunidade.xlsx", rowNames = FALSE)
388   wb <- loadWorkbook("res/tabela da comunidade.xlsx")
389   writeData(wb, sheet = "Sheet 1", x = m_wide_spp)
390   saveWorkbook(wb, "res/tabela da comunidade.xlsx", overwrite = TRUE)
391
392   # Tabela final ajustada no Excel
393   library(readxl)
394   spp_gtable <- read_excel(
395     path = "res/tabela da comunidade.xlsx",

```

```

396   sheet = "tabela_final",
397   col_names = T,
398   range = "A1:AE61") #ajustar para o tamanho da tabela final
399 spp_gtttable
400 library(gt)
401 spp_gtttable <- gt(spp_gtttable)
402 spp_gtttable <- sub_missing(spp_gtttable,
403                             columns = everything(), missing_text = "")
404 #spp_gtttable <- fmt_number(spp_gtttable,
405 #                           columns = "F.O.(%)", decimals = 1)
406 spp_gtttable
407 gtsave(spp_gtttable, "res/gt-spp_gtttable_xlsx.html")
408 saveRDS(spp_gtttable, "res/gt-spp_gtttable_xlsx.rds")

1  # Escolher sumário de uma variavel
2  m
3  var <- "h.roots"
4  m[m$Variable == var, "mean"] #cada valor de var
5  summary(m[m$Variable == var, "mean"]) #sumário dos valores de var
6
7  # Escolher sumário de um grupo de variáveis do df m
8  vars <- unique(grep("^h\\.\\.", m$Variable, value = TRUE))
9  summaries <- list() #cria uma lista vazia para guardar os sumários
10 # Loop para cada variável do grupo e guarda em summaries
11 for (var in vars) {
12   summaries[[var]] <- summary(m[m$Variable == var, "mean"])
13 }
14 #var is a temporary variable used in the for loop to iterate through
15 #each variable name that starts with "h."
16
17 summaries
18 summary_table <- do.call(rbind, lapply(summaries, as.data.frame.list))
19 round(summary_table, 2)
20 #sink(file = "summary_h.txt", split = TRUE)
21 round(summary_table[order(summary_table$Mean, decreasing = FALSE), ], 2)
22 #sink()
23
24 # Tabela limpa
25 #summary_table <- cbind(Variable = rownames(summary_table), summary_table)
26 #rownames(summary_table) <- NULL
27 #colnames(summary_table) <- c("Variable", "Min", "Q1", "Median", "Mean", "Q3", "Max")
28 #summary_table

```

```

1  #ZOOEBIO_part - UPGMA----
2  #####....-----
3
4  ##ORGANIZANDO DADOS----
5
6  dev.off() #apaga os graficos, se houver algum
7  rm(list=ls(all=TRUE)) #limpa a memória
8  cat("\014") #limpa o console
9
10 t_grps <- read.csv("res/t_grps.csv",
11                   sep = ";", dec = ".",
12                   row.names = 1,
13                   header = TRUE,
14                   na.strings = NA)
15 m_dens <- read.csv("res/m_dens.csv",
16                  sep = ";", dec = ".",
17                  row.names = 1,
18                  header = TRUE,
19                  na.strings = NA)
20 m_hab_part <- read.csv("res/m_hab_part.csv",
21                      sep = ";", dec = ".",
22                      row.names = 1,
23                      header = TRUE,
24                      na.strings = NA)
25 m_cont <- read.csv("res/m_cont.csv",
26                   sep = ";", dec = ".",
27                   row.names = 1,
28                   header = TRUE,
29                   na.strings = NA)
30
31 m_dens
32 str(m_dens)
33
34 #SEPARANDO OS PERÍODOS
35
36 library(vegan)
37 # Fazendo a média por ponto
38 ponto_id <- sub("-H[0-9]+$", "", rownames(m_dens))
39 m_dens_media <- aggregate(m_dens, by = list(Ponto = ponto_id), FUN = mean)
40 rownames(m_dens_media) <- m_dens_media$Ponto
41 m_dens_media$Ponto <- NULL
42 # Separar por período
43 seco_med <- m_dens_media[startsWith(rownames(m_dens_media), "S-"), ]

```



```

44 chuv_med <- m_dens_media[startsWith(rownames(m_dens_media), "C-"), ]
45 # Filtrar COLUNAS zeradas (espécies ausentes no período)
46 seco_med <- seco_med[, colSums(seco_med) > 0]
47 chuv_med <- chuv_med[, colSums(chuv_med) > 0]
48 # Filtrar LINHAS zeradas (pontos sem organismos)
49 seco_med <- seco_med[rowSums(seco_med) > 0, ]
50 chuv_med <- chuv_med[rowSums(chuv_med) > 0, ]
51 # Transformação
52 seco_trns <- asin(sqrt(decostand(seco_med,
53                               method="total", MARGIN = 2)))
54 chuv_trns <- asin(sqrt(decostand(chuv_med,
55                               method="total", MARGIN = 2)))
56 # Distâncias de Bray-Curtis e UPGMA
57 dist_seco <- vegdist(seco_trns, method = "bray")
58 dist_chuv <- vegdist(chuv_trns, method = "bray")
59 upgma_seco <- hclust(dist_seco, method = "average")
60 upgma_chuv <- hclust(dist_chuv, method = "average")
61
62 # Plot dos dendrogramas agrupados por ponto
63 upgma_seco$labels <- sub("[A-Z]-", "", upgma_seco$labels)
64 upgma_chuv$labels <- sub("[A-Z]-", "", upgma_chuv$labels)
65 dend_seco <- as.dendrogram(hclust(dist_seco, method = "average"))
66 dend_chuv <- as.dendrogram(hclust(dist_chuv, method = "average"))
67
68 #png("res/fig-upgma.png", width=11, height=9, units="in", res=400)
69 par(mfrow = c(1, 2), mar = c(5, 4, 4, 6))
70 plot(as.dendrogram(upgma_seco),
71      horiz = TRUE,
72      main = "UPGMA - Seco",
73      xlab = "Dissimilaridade (Bray-Curtis)",
74      xlim = c(max(attr(dend_seco, "height")), 0))
75 plot(as.dendrogram(upgma_chuv),
76      horiz = TRUE,
77      main = "UPGMA - Chuvoso",
78      xlab = "Dissimilaridade (Bray-Curtis)",
79      xlim = c(max(attr(dend_chuv, "height")), 0))
80 par(mfrow = c(1, 1))
81 #dev.off()
82
83 # Gerar a matriz de distância cofenética a partir da árvore UPGMA
84 cofe_seco <- cophenetic(upgma_seco)
85 cofe_chuv <- cophenetic(upgma_chuv)
86 # Calcular a correlação com a matriz de distância original (Bray-Curtis)

```

```

87 cor_seco <- cor(dist_seco, cofe_seco)
88 cor_chuv <- cor(dist_chuv, cofe_chuv)
89 # Visualizar os resultados
90 cat("Correlação Cofenética (Seco):", round(cor_seco, 3), "\n")
91 cat("Correlação Cofenética (Chuvoso):", round(cor_chuv, 3), "\n")

```

3.3 Diversity Partitioning

```

1  #ZOOEBIO_part - ADDBPART----
2  #####....-----
3
4  ##ORGANIZANDO DADOS----
5
6  dev.off() #apaga os graficos, se houver algum
7  rm(list=ls(all=TRUE)) #limpa a memória
8  cat("\014") #limpa o console
9
10 m_esc <- read.csv("res/m_esc.csv",
11                  sep = ";", dec = ".",
12                  row.names = 1,
13                  header = TRUE,
14                  na.strings = NA)
15 m_dens <- read.csv("res/m_dens.csv",
16                  sep = ";", dec = ".",
17                  row.names = 1,
18                  header = TRUE,
19                  na.strings = NA)
20 m_cont <- read.csv("res/m_cont.csv",
21                  sep = ";", dec = ".",
22                  row.names = 1,
23                  header = TRUE,
24                  na.strings = NA)
25
26 #BUSCANDO AS ESCALAS
27
28 # Partição total
29 colnames(m_esc)
30 pt.total <- m_esc[, c("pt.rio", "pt.periodo", "pt.rebio")]
31 head(pt.total)
32 dim(pt.total)
33
34 # Partição seco

```

```

35 pt.seco <- m_esc[, c("s.habitat", "s.ponto", "s.rio", "s.rebio")]
36 pt.seco <- pt.seco[grepl("^S-", rownames(pt.seco)), ]
37 head(pt.seco)
38 dim(pt.seco)
39 m_cont_s <- m_cont[grepl("^S-", rownames(m_cont)), ]
40
41 # Partição chuvoso
42 pt.chuva <- m_esc[, c("c.habitat", "c.ponto", "c.rio", "c.rebio")]
43 pt.chuva <- pt.chuva[grepl("^C-", rownames(pt.chuva)), ]
44 head(pt.chuva)
45 dim(pt.chuva)
46 m_cont_c <- m_cont[grepl("^C-", rownames(m_cont)), ]
47
48 # Partição bb.seco (A)
49 bb.seco <- m_esc[, c("bb.habitat", "bb.ponto", "bb.rio")]
50 bb.seco <- bb.seco[grepl("S-B-", rownames(bb.seco)), ]
51 head(bb.seco)
52 dim(bb.seco)
53 m_cont_bb.seco <- m_cont[grepl("S-B-", rownames(m_cont)), ]
54
55 # Partição bb.chuva (B)
56 bb.chuva <- m_esc[, c("bb.habitat", "bb.ponto", "bb.rio")]
57 bb.chuva <- bb.chuva[grepl("C-B-", rownames(bb.chuva)), ]
58 head(bb.chuva)
59 dim(bb.chuva)
60 m_cont_bb.chuva <- m_cont[grepl("C-B-", rownames(bb.chuva)), ]
61
62 # Partição ca.seco (C)
63 ca.seco <- m_esc[, c("ca.habitat", "ca.ponto", "ca.rio")]
64 ca.seco <- ca.seco[grepl("S-C-", rownames(ca.seco)), ]
65 head(ca.seco)
66 dim(ca.seco)
67 m_cont_ca.seco <- m_cont[grepl("S-C-", rownames(ca.seco)), ]
68
69 # Partição ca.chuva (D)
70 ca.chuva <- m_esc[, c("ca.habitat", "ca.ponto", "ca.rio")]
71 ca.chuva <- ca.chuva[grepl("C-C-", rownames(ca.chuva)), ]
72 head(ca.chuva)
73 dim(ca.chuva)
74 m_cont_ca.chuva <- m_cont[grepl("C-C-", rownames(ca.chuva)), ]
75
76 #PARTIÇÃO
77

```

```

78  #pt.seco (m_cont_s), pt.chuva (m_cont_c)
79  #bb.seco(A) (m_cont_bb.seco), bb.chuva(B) (m_cont_bb.chuva)
80  #ca.seco(C) (m_cont_ca.seco), ca.chuva(D) (m_cont_ca.chuva)
81
82  part1 <- ca.chuva
83  cont <- m_cont_ca.chuva
84
85  library(vegan)
86
87  sink(file = "res/part-ca.chuva.txt")
88  #VALORES ABSOLUTOS
89  part.a <- adipart(y = cont, x = part1, index = "richness", relative = FALSE,
90                  weights = "unif", nsimul = 999)
91  part.a
92
93  #EM PORCENTAGEM
94  part.p <- adipart(y = cont, x = part1, index = "richness", relative = TRUE,
95                  weights = "unif", nsimul = 999)
96  part.p
97
98  #str(part.p)
99
100 # Extrair valores do objeto part.p
101
102 part2 <- part.p
103
104 statistic <- part2$statistic
105 SES <- part2$oecosimu$z
106 mean <- part2$oecosimu$means
107 lower_2.5 <- apply(part2$oecosimu$simulated, 1, quantile, probs = 0.025)
108 median_50 <- apply(part2$oecosimu$simulated, 1, quantile, probs = 0.50)
109 upper_97.5 <- apply(part2$oecosimu$simulated, 1, quantile, probs = 0.975)
110 Pr_sim <- part2$oecosimu$pval
111
112 # Definir função para adicionar significância
113 get_significance <- function(p) {
114   if (p <= 0.001) {
115     return("****")
116   } else if (p <= 0.01) {
117     return("***")
118   } else if (p <= 0.05) {
119     return("**")
120   } else if (p <= 0.1) {

```

```

121     return(".")
122   } else {
123     return(" ")
124   }
125 }
126 significance <- sapply(Pr_sim, get_significance)
127
128 # Criar dataframe com todas as informações
129
130 diversity_data <- data.frame(
131   Component = names(statistic),
132   Observed = round(statistic, 2),
133   SES = round(SES, 2),
134   Mean = round(mean, 2),
135   Lower_2.5 = round(lower_2.5, 2),
136   Median_50 = round(median_50, 2),
137   Upper_97.5 = round(upper_97.5, 2),
138   Pr_sim = Pr_sim,
139   Sign = significance
140 )
141
142 # Exibir a tabela
143 library(gt)
144 print(diversity_data)
145 #str(part2)
146 gt(diversity_data)
147
148 # Criar dataframe com valores observados e esperados
149 observed_values <- part2$statistic
150 observed_values
151 expected_values <- part2$oecosimu$means
152 expected_values
153
154 # Combine observed and expected values into a dataframe
155 df <- data.frame(
156   Component = rep(names(observed_values), 2), # Use names from observed_values
157   Type = rep(c("Observed", "Expected"), each = length(observed_values)),
158   Value = c(observed_values, expected_values) # Observado primeiro, Esperado depois
159 )
160
161 df$Type <- factor(df$Type, levels = c("Observed", "Expected"))
162 df
163 df_filtered <- df[!df$Component %in% c("alpha.2", "alpha.3", "gamma"), ]

```

```

164 df_filtered$Component <- factor(df_filtered$Component, levels = c("beta.3", "beta.2", "beta.1"))
165 df_filtered
166
167 diversity_data <- df_filtered
168 diversity_data
169
170 # Calcular porcentagens para cada componente
171 diversity_data <- diversity_data %>%
172   group_by(Type) %>%
173   mutate(Percentage = Value / sum(Value) * 100)
174
175 hier <- hiersimu(y = cont, x = part1, FUN = diversity)
176 hier
177 sink()
178
179 #GRÁFICOS
180
181 library(ggplot2)
182 library(dplyr)
183
184 # Partição total
185
186 ggplot_pt.total <- ggplot(diversity_data, aes(fill = Component, y = Percentage, x = Type)) +
187   geom_bar(position = "stack", stat = "identity") +
188   labs(title = "Total",
189        y = "% of the total diversity",
190        x = "") +
191   scale_fill_manual(values = c("alpha.1" = "black",
192                                "beta.1" = "darkgray",
193                                "beta.2" = "gray",
194                                "beta.3" = "lightgray")) +
195   theme_minimal() +
196   theme(
197     axis.text.x = element_text(angle = 0, hjust = 0.5),
198     panel.grid = element_blank(),
199   )
200 ggplot_pt.total
201 ggsave(ggplot_pt.total, dpi = 300, filename = "res/fig-part_total.png")
202
203 # Partição seco
204
205 ggplot_pt.seco <- ggplot(diversity_data, aes(fill = Component, y = Percentage, x = Type)) +
206   geom_bar(position = "stack", stat = "identity") +

```

```

207   labs(title = "Seco",
208         y = "% of the total diversity",
209         x = "") +
210   scale_fill_manual(values = c("alpha.1" = "black",
211                                "beta.1" = "darkgray",
212                                "beta.2" = "gray",
213                                "beta.3" = "lightgray")) +
214   theme_minimal() +
215   theme(
216     axis.text.x = element_text(angle = 0, hjust = 0.5),
217     panel.grid = element_blank(),
218   )
219 ggplot_pt.seco
220 ggsave(ggplot_pt.total, dpi = 300, filename = "res/fig-part_seco.png")
221
222 # Partição chuvoso
223
224 ggplot_pt.chuva <- ggplot(diversity_data, aes(fill = Component, y = Percentage, x = Type)) +
225   geom_bar(position = "stack", stat = "identity") +
226   labs(title = "Chuvoso",
227         y = "% of the total diversity",
228         x = "") +
229   scale_fill_manual(values = c("alpha.1" = "black",
230                                "beta.1" = "darkgray",
231                                "beta.2" = "gray",
232                                "beta.3" = "lightgray")) +
233   theme_minimal() +
234   theme(
235     axis.text.x = element_text(angle = 0, hjust = 0.5),
236     panel.grid = element_blank(),
237   )
238 ggplot_pt.chuva
239 ggsave(ggplot_pt.total, dpi = 300, filename = "res/fig-part_chuva.png")
240
241 # bb.seco (A)
242
243 ggplot_bb.seco <- ggplot(diversity_data, aes(fill = Component, y = Percentage, x = Type)) +
244   geom_bar(position = "stack", stat = "identity") +
245   labs(title = "(A)",
246         y = "% of the total diversity",
247         x = "") +
248   scale_fill_manual(values = c("alpha.1" = "black",
249                                "beta.1" = "darkgray",

```

```

250         "beta.2" = "gray",
251         "beta.3" = "lightgray")) +
252     theme_minimal() +
253     theme(
254         axis.text.x = element_text(angle = 0, hjust = 0.5),
255         panel.grid = element_blank(),
256     )
257 ggplot_bb.seco
258 ggsave(ggplot_bb.seco, dpi = 300, filename = "res/fig-bb.seco.png")
259
260 # bb.chuva (B)
261
262 ggplot_bb.chuva <- ggplot(diversity_data, aes(fill = Component, y = Percentage, x = Type)) +
263     geom_bar(position = "stack", stat = "identity") +
264     labs(title = "(B)",
265          y = "% of the total diversity",
266          x = "") +
267     scale_fill_manual(values = c("alpha.1" = "black",
268                                  "beta.1" = "darkgray",
269                                  "beta.2" = "gray",
270                                  "beta.3" = "lightgray")) +
271     theme_minimal() +
272     theme(
273         axis.text.x = element_text(angle = 0, hjust = 0.5),
274         panel.grid = element_blank(),
275     )
276 ggplot_bb.chuva
277 ggsave(ggplot_bb.chuva, dpi = 300, filename = "res/fig-bb.chuva.png")
278
279 # ca.seco (C)
280
281 ggplot_ca.seco <- ggplot(diversity_data, aes(fill = Component, y = Percentage, x = Type)) +
282     geom_bar(position = "stack", stat = "identity") +
283     labs(title = "(C)",
284          y = "% of the total diversity",
285          x = "") +
286     scale_fill_manual(values = c("alpha.1" = "black",
287                                  "beta.1" = "darkgray",
288                                  "beta.2" = "gray",
289                                  "beta.3" = "lightgray")) +
290     theme_minimal() +
291     theme(
292         axis.text.x = element_text(angle = 0, hjust = 0.5),

```



```

293     panel.grid = element_blank(),
294   )
295   ggplot_ca.seco
296   ggsave(ggplot_ca.seco, dpi = 300, filename = "res/fig-ca.seco.png")
297
298   # ca.chuva (D)
299
300   ggplot_ca.chuva <- ggplot(diversity_data, aes(fill = Component, y = Percentage, x = Type)) +
301     geom_bar(position = "stack", stat = "identity") +
302     labs(title = "(D)",
303          y = "% of the total diversity",
304          x = "") +
305     scale_fill_manual(values = c("alpha.1" = "black",
306                                  "beta.1" = "darkgray",
307                                  "beta.2" = "gray",
308                                  "beta.3" = "lightgray")) +
309     theme_minimal() +
310     theme(
311       axis.text.x = element_text(angle = 0, hjust = 0.5),
312       panel.grid = element_blank(),
313     )
314   ggplot_ca.chuva
315   ggsave(ggplot_ca.chuva, dpi = 300, filename = "res/fig-ca.chuva.png")
316
317   # PAINEL DE GRÁFICOS
318
319   library(gridExtra)
320
321   # 1. Organizar os gráficos em 2 linhas e 2 colunas
322
323   painel_completo <- grid.arrange(
324     ggplot_bb.seco, ggplot_bb.chuva,
325     ggplot_ca.seco, ggplot_ca.chuva,
326     ncol = 2, nrow = 2
327   )
328
329   # 2. Salvar o painel em alta resolução (300 DPI)
330   ggsave(
331     filename = "res/fig-painel_partição.png",
332     plot = painel_completo,
333     width = 10,
334     height = 8,
335     dpi = 300

```

```

336 )

1  #TABELA DE PARTIÇÃO----
2  #Feita no Gemini, conferir
3
4  # -----
5  # 1. FUNÇÃO DE EXTRAÇÃO
6  # -----
7  extrair_linha_particao <- function(part_obj, sistema, periodo) {
8
9      mapa_componentes <- c(
10         "alpha.1" = " 1 (micro-habitat)",
11         "alpha.2" = " 2 (ponto)",
12         "alpha.3" = " 3 (riacho)",
13         "beta.1"  = " 1 (entre micro-habitats)",
14         "beta.2"  = " 2 (entre pontos)",
15         "beta.3"  = " 3 (entre riachos)",
16         "gamma"   = " (Riqueza total)"
17     )
18
19     stat <- part_obj$statistic
20     comp_names <- names(stat)
21
22     df <- data.frame(
23         Sistema      = sistema,
24         Período      = periodo,
25         Componente   = ifelse(comp_names %in% names(mapa_componentes),
26                               mapa_componentes[comp_names], comp_names),
27         Observado    = sprintf("%.2f%%", stat * 100),
28         Esperado     = sprintf("%.2f%%", part_obj$oecosimu$means * 100),
29         SES          = sprintf("%.2f", part_obj$oecosimu$z),
30         p_val        = part_obj$oecosimu$pval,
31         stringsAsFactors = FALSE
32     )
33
34     df$p_valor <- ifelse(df$p_val <= 0.001, "< 0.001*",
35                         ifelse(df$p_val <= 0.05, sprintf("%.3f*", df$p_val),
36                               sprintf("%.3f", df$p_val)))
37
38     return(df %>% select(-p_val))
39 }
40
41 # -----

```

```

42 # 2. PROCESSAMENTO E RUNDOWN DOS OBJETOS ADIPART
43 # -----
44 # Rodando o adipart para cada cenário e gerando os objetos corretos
45
46 p_ca_seco <- adipart(y = m_cont_ca.seco, x = ca.seco, index = "richness", relative = TRUE)
47 p_ca_chuva <- adipart(y = m_cont_ca.chuva, x = ca.chuva, index = "richness", relative = TRUE)
48
49 p_bb_seco <- adipart(y = m_cont_bb.seco, x = bb.seco, index = "richness", relative = TRUE)
50 p_bb_chuva <- adipart(y = m_cont_bb.chuva, x = bb.chuva, index = "richness", relative = TRUE)
51
52 p_pt_seco <- adipart(y = m_cont_s, x = pt.seco, index = "richness", relative = TRUE)
53 p_pt_chuva <- adipart(y = m_cont_c, x = pt.chuva, index = "richness", relative = TRUE)
54
55 # -----
56 # 3. CONSOLIDAÇÃO DA TABELA 1
57 # -----
58 tabela_1_raw <- rbind(
59   extrair_linha_particao(p_ca_seco, "Riacho Caiana (CA)", "Seco"),
60   extrair_linha_particao(p_ca_chuva, "Riacho Caiana (CA)", "Chuvoso"),
61   extrair_linha_particao(p_bb_seco, "Riacho Barro Branco (BB)", "Seco"),
62   extrair_linha_particao(p_bb_chuva, "Riacho Barro Branco (BB)", "Chuvoso"),
63   extrair_linha_particao(p_pt_seco, "Partição Total (PT)", "Seco"),
64   extrair_linha_particao(p_pt_chuva, "Partição Total (PT)", "Chuvoso")
65 )
66
67 # -----
68 # 4. IMPRESSÃO DA TABELA GT
69 # -----
70 tabela_1_final <- tabela_1_raw %>%
71   gt(groupname_col = "Sistema") %>%
72   tab_header(
73     title = md("**Tabela 1.** Partição aditiva da riqueza de espécies zooplanctônicas nas di
74   ) %>%
75   cols_label(
76     Período = "Período",
77     Componente = "Componente Hierárquico",
78     Observado = "Observado (%)",
79     Esperado = "Esperado (%)",
80     SES = "SES",
81     p_valor = "p-valor"
82   ) %>%
83   cols_align(
84     align = "center",

```

```

85     columns = c("Período", "Observado", "Esperado", "SES", "p_valor")
86   )
87   tabela_1_final
88
89   # Formatar nomes das colunas diretamente
90   tabela_1_simples <- tabela_1_raw
91   colnames(tabela_1_simples) <- c("Sistema", "Período", "Componente", "Obs (%)", "Esp (%)", "SI
92
93   # Imprimir no console
94   print(tabela_1_simples, row.names = FALSE)
95
96   # Exportando dados para Excel
97   library(openxlsx)
98   #write.xlsx(tabela_1_simples, file = "res/tabela de partição.xlsx", rowNames = FALSE)
99   wb <- loadWorkbook("res/tabela de partição.xlsx")
100  writeData(wb, sheet = "Sheet 1", x = tabela_1_simples)
101  saveWorkbook(wb, "res/tabela de partição.xlsx", overwrite = TRUE)
102
103  # Tabela final ajustada no Excel
104  library(readxl)
105  part_gtttable <- read_excel(
106    path = "res/tabela de partição.xlsx",
107    sheet = "tabela_final",
108    range = "A1:G35") #ajustar para o tamanho da tabela final
109  part_gtttable
110  library(gt)
111  part_gtttable <- gt(part_gtttable)
112  part_gtttable <- sub_missing(part_gtttable,
113                                columns = everything(), missing_text = "")
114  part_gtttable
115  gtsave(part_gtttable, "res/gt-part_gtttable.xlsx.html")
116  saveRDS(part_gtttable, "res/gt-part_gtttable.xlsx.rds")

```

4 Discussion

Conclusions

Acknowledgements

We are greatly indebted to the staff of the REBIO Guaribas for field work assistance. LCA is grateful to Programa de Pós-Graduação em Ecologia e Conservação (PPGEC-UEPB) and CAPES for scholarship granted. EGS, MMSM and MVBS are grateful to Programa de Iniciação Científica (PIBIC-UEPB) and CNPq for scholarship granted.

Authorship contribution statement

Authorship of this paper is based on CRediT (2026).

Ellen Gomes da Silva: Data curation, Formal analysis, Investigation, Software, Visualization, Writing – original draft.

Mayara Mirelly da Silva Monteiro: Data curation, Formal analysis, Investigation.

Ludmilla Cavalcanti Antunes Lucena: Supervision, Validation.

Thais Xavier de Melo: Supervision, Validation.

Elvio Sergio Figueredo Medeiros: Conceptualization, Data Curation, Formal Analysis, Funding Acquisition, Investigation, Methodology, Project Administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Ethical approval

Activities inside the REBIO Guaribas were authorized by the Instituto Chico Mendes de Conservação da Biodiversidade (license no. SISBIO-89415-1, SISBIO-89429-2 and SISBIO-89415-2).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this manuscript, the authors used ChatGPT (OpenAI) to assist with English language editing and improve the clarity and fluency of the writing. No AI was used for data analysis, interpretation, or generation of scientific conclusions. The authors carefully reviewed and revised all AI-generated suggestions and take full responsibility for the accuracy and content of the final manuscript (CNPq, 2026).

References

- APHA. (1998). *Standard methods for the examination of water and wastewater*. <https://www.standardmethods.org/>
- Arruda, D. B., Cunha, B. P., & Rêgo, K. M. C. (2013). Conflitos entre REBIO Guaribas e comunidades locais: (in)justiça ambiental e ecologia política. *Revista Direitos Emergentes na Sociedade Global*, 2(2), 280–304. <https://doi.org/10.5902/2316305410852>
- Ávila, A. C., Stenert, C., & Maltchik, L. (2011). Partitioning macroinvertebrate diversity across different spatial scales in southern brazil coastal wetlands. *Wetlands*, 31(3), 459–469. <https://doi.org/10.1007/s13157-011-0178-3>
- Baselga, A. (2010). Partitioning the turnover and nestedness components of beta diversity. *Global Ecology and Biogeography*, 19(1), 134–143. <https://doi.org/10.1111/j.1466-8238.2009.00490.x>
- Baselga, A. (2012). The relationship between species replacement, dissimilarity derived from nestedness, and nestedness. *Global Ecology and Biogeography*, 21(12), 1223–1232. <https://doi.org/10.1111/j.1466-8238.2011.00756.x>
- Biondini, M. E., Bonham, C. D., & Redente, E. F. (1985). Secondary successional patterns in a sagebrush (*artemisia tridentata*) community as they relate to soil disturbance and soil biological activity. *Vegetatio*, 60(1), 25–36. <https://doi.org/10.1007/BF00053909>
- Bowen, K. L. (2017). *Methods for the determination of zooplankton density, biomass and secondary production* (p. 50). <https://publications.gc.ca/site/eng/9.836661/publication.html>
- Brasil. (2000). *Lei no 9.985, de 18 de julho de 2000. Regulamenta o art. 225, § 1o, incisos i, II, III e VII da constituição federal, institui o sistema nacional de unidades de conservação da natureza e dá outras providências*. https://www.planalto.gov.br/ccivil_03/leis/19985.htm
- Chao, A., Gotelli, N. J., Hsieh, T. C., Sander, E. L., Ma, K. H., Colwell, R. K., & Ellison, A. M. (2014). Rarefaction and extrapolation with hill numbers: A framework for sampling and estimation in species diversity studies. *Ecological Monographs*, 84(1), 45–67. <https://doi.org/10.1890/13-0133.1>
- CNPq. (2026). *Portaria 2664/2026 de 6 de março de 2026 - política de integridade na atividade científica do CNPq*. http://memoria2.cnpq.br/web/guest/view/-/journal_content/56_INSTANCE_0oED/10157/23142775

- CReditT. (2026). *CReditT - contributor role taxonomy*. <https://credit.niso.org/>
- Crist, T. O., Veech, J. A., Gering, J. C., & Summerville, K. S. (2003). Partitioning species diversity across landscapes and regions: A hierarchical analysis of α , β , and γ diversity. *The American Naturalist*, 162(6), 734–743. <https://doi.org/10.1086/378901>
- Gering, J. C., Crist, T. O., & Veech, J. A. (2003). Additive partitioning of species diversity across multiple spatial scales: Implications for regional conservation of biodiversity. *Conservation Biology*, 17(2), 488–499. <https://doi.org/10.1046/j.1523-1739.2003.01465.x>
- Gouveia, R. S. D., Lira, G. L. D. A., Ramos, T. P. A., & Medeiros, E. S. F. (2017). Ichthyofauna of the reserva biológica guaribas and surrounding areas, state of paraíba, brazil. *Check List*, 13(5), 581–590. <https://doi.org/10.15560/13.5.581>
- Haney, J. F., & Hall, D. J. (1973). Sugar-coated *Daphnia* : A preservation technique for Cladocera1. *Limnology and Oceanography*, 18(2), 331–333. <https://doi.org/10.4319/lo.1973.18.2.0331>
- Heinrichs, R., Otto, R., Magalhães, A., & Meirelles, G. C. (2017). *Importance of sugarcane in brazilian and world bioeconomy* (S. Dabbert, I. Lewandowski, J. Weiss, & A. Pyka, Eds.; pp. 205–217). Springer International Publishing. https://doi.org/10.1007/978-3-319-58374-7_11
- Lê, S., Josse, J., & Husson, F. (2008). FactoMineR: An r package for multivariate analysis. *Journal of Statistical Software*, 25(1), 1–18.
- Legendre, P., & Legendre, L. (1998). *Numerical ecology* (2nd English ed).
- Ligeiro, R., Melo, A. S., & Callisto, M. (2010). Spatial scale and the diversity of macroinvertebrates in a neotropical catchment. *Freshwater Biology*, 2010(55), 424–435. <https://doi.org/10.1111/j.1365-2427.2009.02291.x>
- Mackereth, F. J. H., Heron, J., & Talling, J. F. (1978). Water analysis: Some revised methods for limnologists.(scientific publication n. 36) freshwater biological association. *Cumbria and Dorset, England*.
- Magurran, A. E. (2013). *Measuring biological diversity*. Wiley-Blackwell.
- Maitland, P. S. (1990). *Biology of fresh waters* (1st ed.). Springer Netherlands. <https://doi.org/10.1007/978-94-011-7852-5>
- McCune, B., & Grace, J. B. (2002). *Analysis of ecological communities*. MjM Software Design.
- Medeiros, E. S. F., Silva, M. J., & Ramos, R. T. C. (2008). Application of catchment- and local-scale variables for aquatic habitat characterization and assessment in the Brazilian semi-arid region. *Neotropical Biology and Conservation*, 3, 13–20. <https://doi.org/10.4013/5440>
- Mehla, M. K., Jeong, J., Cho, G., Wolfe, J. E., & Kan, E. (2026). How agricultural field management modulates downstream transport of antibiotics, bacteria and nutrients in runoff water using edge-of-field monitoring. *Water Research*, 307, 126704. <https://doi.org/10.1016/j.watres.2026.126704>
- Miranda, J. (2012). Ameaças aos peixes de riachos da mata atlântica. *Natureza On Line*, 10, 136–139.
- MMA. (2003). *Plano de Manejo da Reserva Biológica Guaribas*.
- MMA. (2026). *Plano de manejo integrado do fogo 2026-2030. Reserva biológica guaribas* (p. 56). https://www.gov.br/icmbio/pt-br/centrais-de-conteudo/publicacoes/planos-de-manejo-integrado-do-fogo/Plano_de_Manejo_Integrado_do_Fogo_da_Reserva

- Mugodo, J., Kennard, M. J., Liston, P., Nichols, S., Linke, S., Norris, R. H., & Lintermans, M. (2006). Local stream habitat variables predicted from catchment scale characteristics are useful for predicting fish distribution. *Hydrobiologia*, 572(1), 59–70. <https://doi.org/10.1007/s10750-006-0252-7>
- Oksanen, J., Simpson, G. L., Blanchet, F. G., Kindt, R., Legendre, P., Minchin, P. R., O'Hara, R. B., Solymos, P., Stevens, M. H. H., Szoecs, E., Wagner, H., Barbour, M., Bedward, M., Bolker, B., Borcard, D., Borman, T., Carvalho, G., Chirico, M., Caceres, M., ... Weedon, J. (n.d.). *Vegan: Community ecology package*. <https://doi.org/10.32614/CRAN.package.vegan>
- Peel, M. C., Finlayson, B. L., & McMahon, T. A. (2007). Updated world map of the köppen-geiger climate classification. *Hydrology and Earth System Sciences*, 11(5), 1633–1644. <https://doi.org/10.5194/hess-11-1633-2007>
- Pereira, K. M. G., Cordeiro, N. G., Suzuki, L. P. Z. L., & Berg, E. van den. (2026). How important are small fragments for restoration in the central portion of the atlantic forest? *Biological Conservation*, 321, Article 111958. <https://doi.org/https://doi.org/10.1016/j.biocon.2026.111958>
- Pusey, B., Kennard, M. J., & Arthington, A. (2004). *Study area, data collection, analysis and presentation* (B. Pusey, M. J. Kennard, & A. Arthington, Eds.; pp. 26–48). CSIRO Publishing.
- Quinn, G. P., & Keough, M. J. (2023). *Experimental design and data analysis for biologists* (2nd ed.). Cambridge University Press. <https://doi.org/10.1017/9781139568173>
- R Core Team. (2017). *R: A language and environment for statistical computing*. <https://www.R-project.org/>
- Rocha, L. G., Medeiros, E. S. F., & Andrade, H. T. A. (2012). Influence of flow variability on macroinvertebrate assemblages in an intermittent stream of semi-arid brazil. *Journal of Arid Environments*, 85, 33–40. <https://doi.org/10.1016/j.jaridenv.2012.04.001>
- Shell, T. M., & Collier, K. J. (2018). Partitioning of macroinvertebrate communities in a large new zealand river highlights the role of multiple shore-zone habitat types. *River Research and Applications*, 34(8), 993–1002. <https://doi.org/10.1002/rra.3344>
- Silva, J. M. C., & Casteleti, C. H. M. (2003). *Status of the biodiversity of the atlantic forest of brazil* (C. Galindo-Leal & I. G. Câmara, Eds.; pp. 43–59). CABS; Island Press.
- Silva, J. M. C., Barbosa, L. C. F., Leal, I. R., & Tabarelli, M. (2017). *The caatinga: Understanding the challenges* (J. M. Da Silva, I. Leal, & M. Tabarelli, Eds.; pp. 3–19). Springer.
- Sokal, R. R., & Rohlf, F. J. (1995). *Biometry: The principles and practice of statistics in biological research* (3rd ed.). W.H. Freeman; Company.
- Strahler, A. N. (1957). Quantitative analysis of watershed geomorphology. *Eos, Transactions American Geophysical Union*, 38(6), 913–920. <https://doi.org/10.1029/TR038i006p00913>
- Veech, J. A., Crist, T. O., & Summerville, K. S. (2003). Intraspecific aggregation decreases local species diversity of arthropods. *Ecology*, 84(12), 3376–3383. <https://doi.org/10.1890/02-3127>
- Zar, J. H. (2010). *Biostatistical analysis* (5th ed.). Prentice Hall.

Figures and Tables

Figure 1: Location of sampling sites along the Barro Branco Stream (Mamanguape, PB) (Site 1 06°43'06"S and 35°10'54"W; Site 2 06°42'38"S and 35°10'38"W; Site 3 06°43'06"S and 35°10'54"W; Site 4 06°41'46"S and 35°10'36"W; Site 5 06°40'56"S and 35°10'27"; Site 6 06°40'18"S and 35°10'34"W).

Figure 2: Principal Component Analysis (PCA) biplot of the sampling units (SUs) and environmental variables in the Barro Branco and Caiana streams during the study period, categorized by site type. Points represent the SUs, whereas vectors represent the environmental variables associated with habitat structure. Ellipses represent 95% confidence regions for the distribution of the groups of points, calculated based on the t-distribution.

Figure 3: Principal Component Analysis (PCA) biplot of the sampling units (SUs) and environmental variables in the Barro Branco and Caiana streams during the study period, categorized by stream. Points represent the SUs, whereas vectors represent the environmental variables associated with habitat structure. Ellipses represent 95% confidence regions for the distribution of the groups of points, calculated based on the t-distribution.

Figure 4: Principal Component Analysis (PCA) biplot of the sampling units (SUs) and environmental variables in the Barro Branco and Caiana streams during the study period, categorized by season. Points represent the SUs, whereas vectors represent the environmental variables associated with habitat structure. Ellipses represent 95% confidence regions for the distribution of the groups of points, calculated based on the t-distribution.

Figure 5: Individual-based extrapolation and rarefaction curves for the richness of resting stages of zooplanktonic taxa in each study stream. Shaded areas indicate 95% confidence intervals.

Figure 6: Classification analysis (UPGMA using Bray-Curtis dissimilarity) of wetland macroinvertebrate samples obtained in a. permanent and b. intermitent wetlands. The first number of the sample label indicates the wetland. The second number indicates the habitat type (1=hydrophytes; 2=reed-like emergent plants; and 3=leafy emergent plants). The third number indicates the individual sweeps realized in each habitat (1–3)UPGMA Hierarchical flexible beta clustering ($\alpha = -0.25$) based on Bray-Curtis distance of (a) macroinvertebrates and (b) Chironomidae in the Paraíba River during the 2015 wet season. Tag codes indicate different sections of the river: U = upper segment, M = middle segment, and T = Taperoá River segment. R = reaches (1–3), S = sites (1–3), and Hab = habitat types (1–3). Percent chaining <4%.

Figure 7: Observed and expected diversity, partitioned into one alpha and three beta components, expressed as percent of total species richness. Numbers indicate the proportion of randomized samples containing more species than the observed sample for each partition. 1: subsample species richness, 1: beta diversity among sub-samples; 2: beta diversity among pools or habitats within pool, and 3: beta diversity among river reaches.

Table 1: Environmental variables averaged for sampling sites (minimum-maximum) for the Barro Branco stream (Mamanguape, PB).

Table 2: Nutrient concentration (g L⁻¹) averaged for sampling sites (minimum-maximum) for the Barro Branco stream (Mamanguape, PB).

Table 3: Density (ind/m²) and frequency of occurrence (FO%) of zooplankton collected during the 2024 hydrological cycle.

Table 4: Contribution (expressed in percentage values) of each spatial scale assessed to the total diversity (richness) of the dormant stages of the zooplankton. Numbers in ‘P value’ column indicate the proportion of randomized samples containing more species than the observed sample for each partition.

Figures

Figure 1

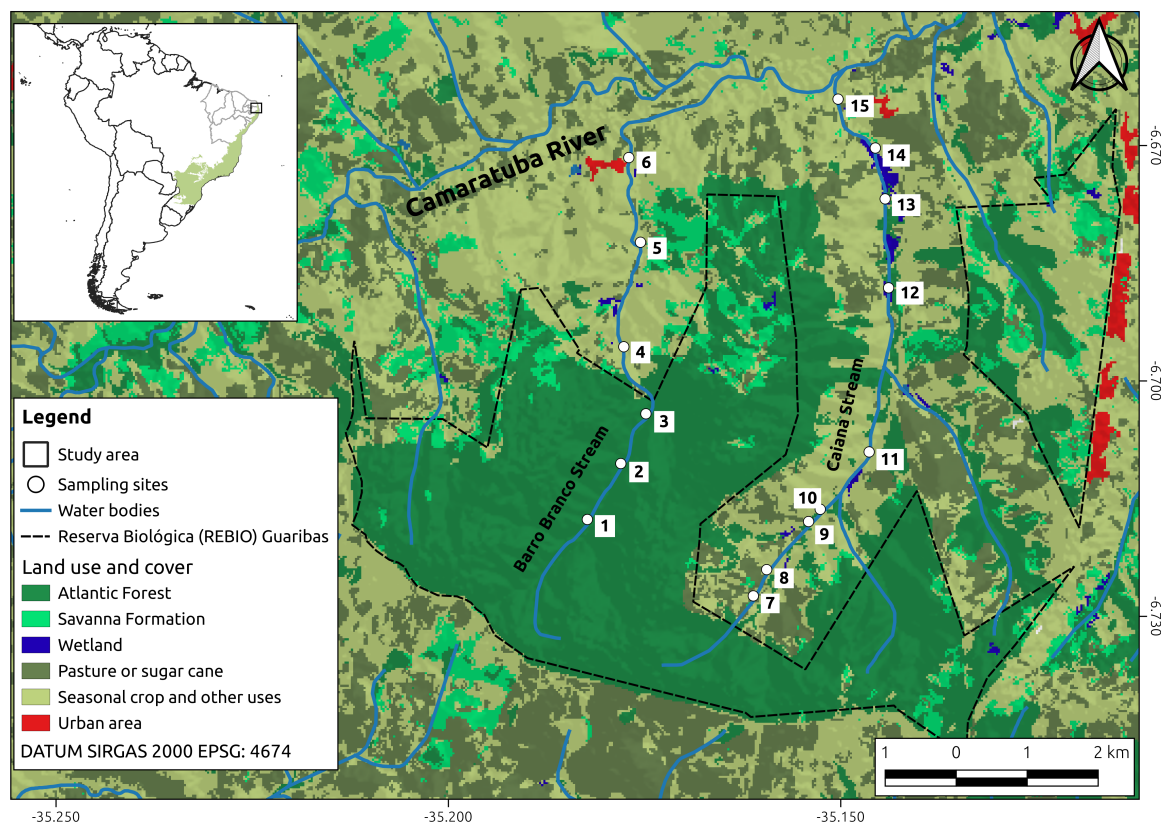


Figure 1

Figure 2

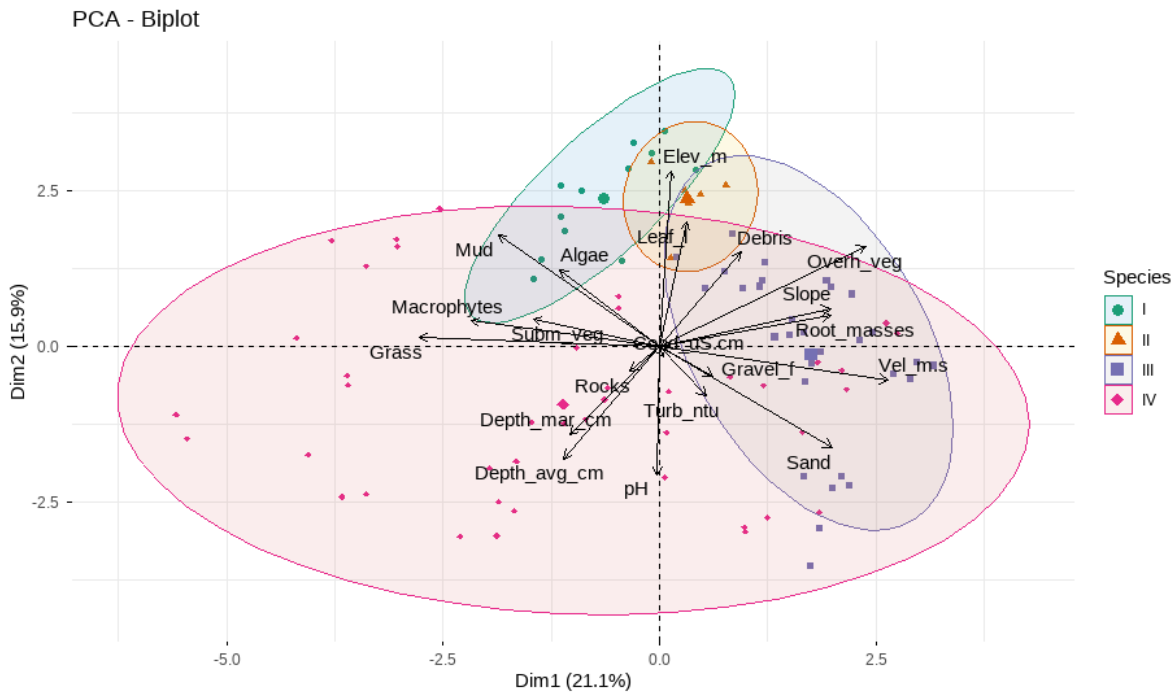


Figure 2

Figure 3

Figure 4

Figure 5

Figure 6

Figure 7

Tables

Table 1

Table 2

Table 3

Table 4

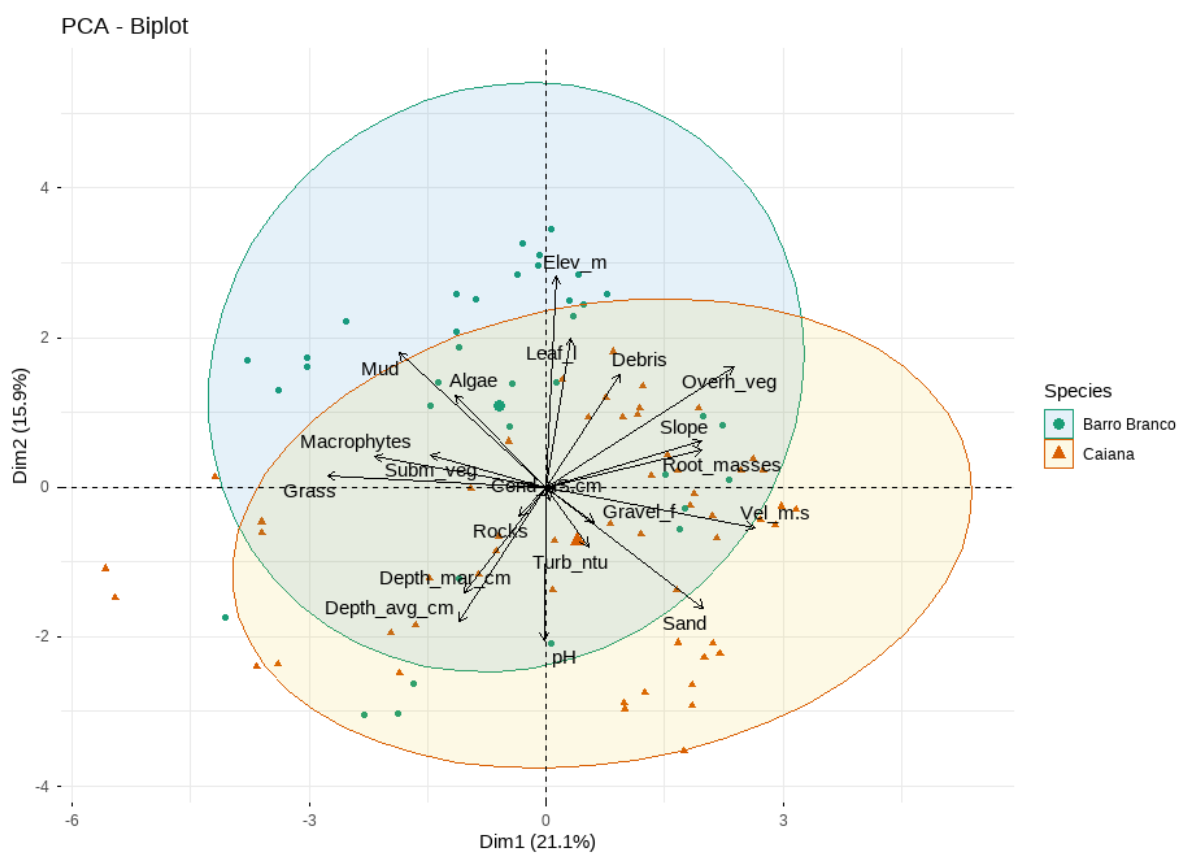


Figure 3

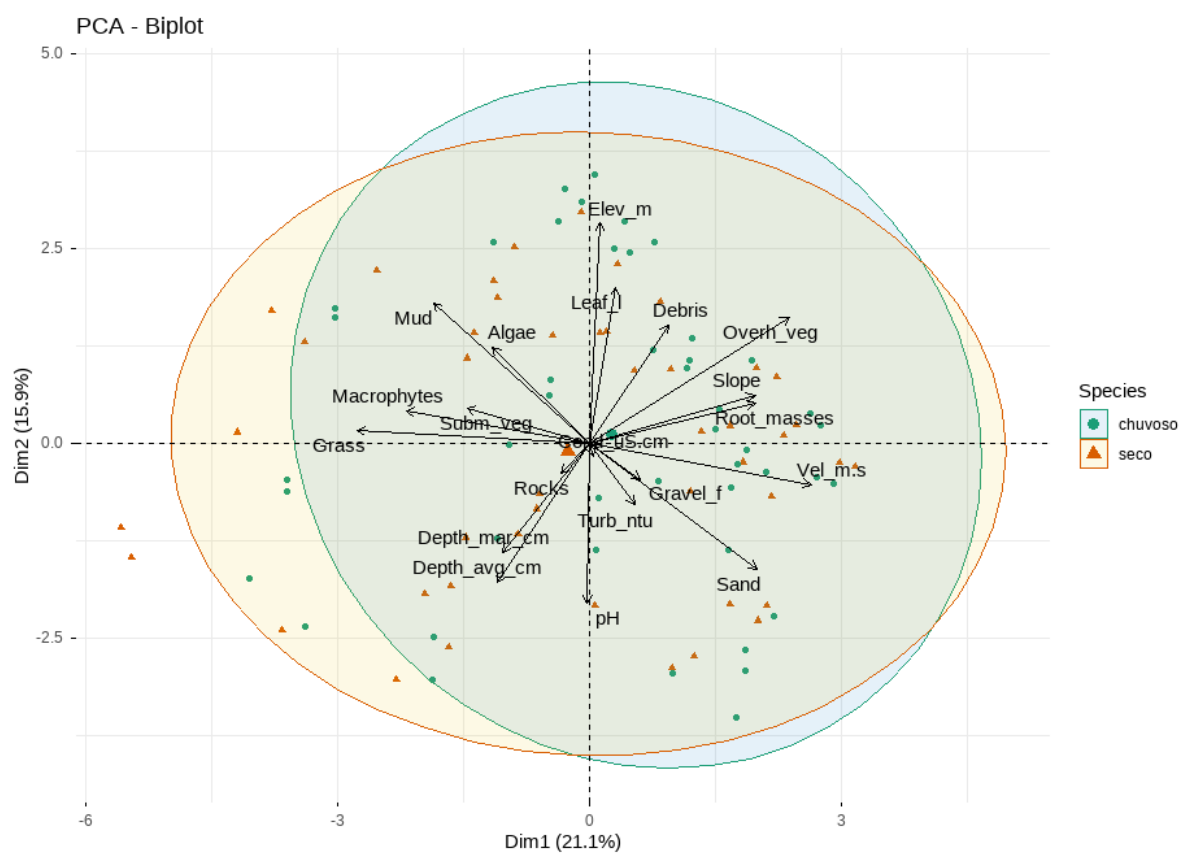


Figure 4

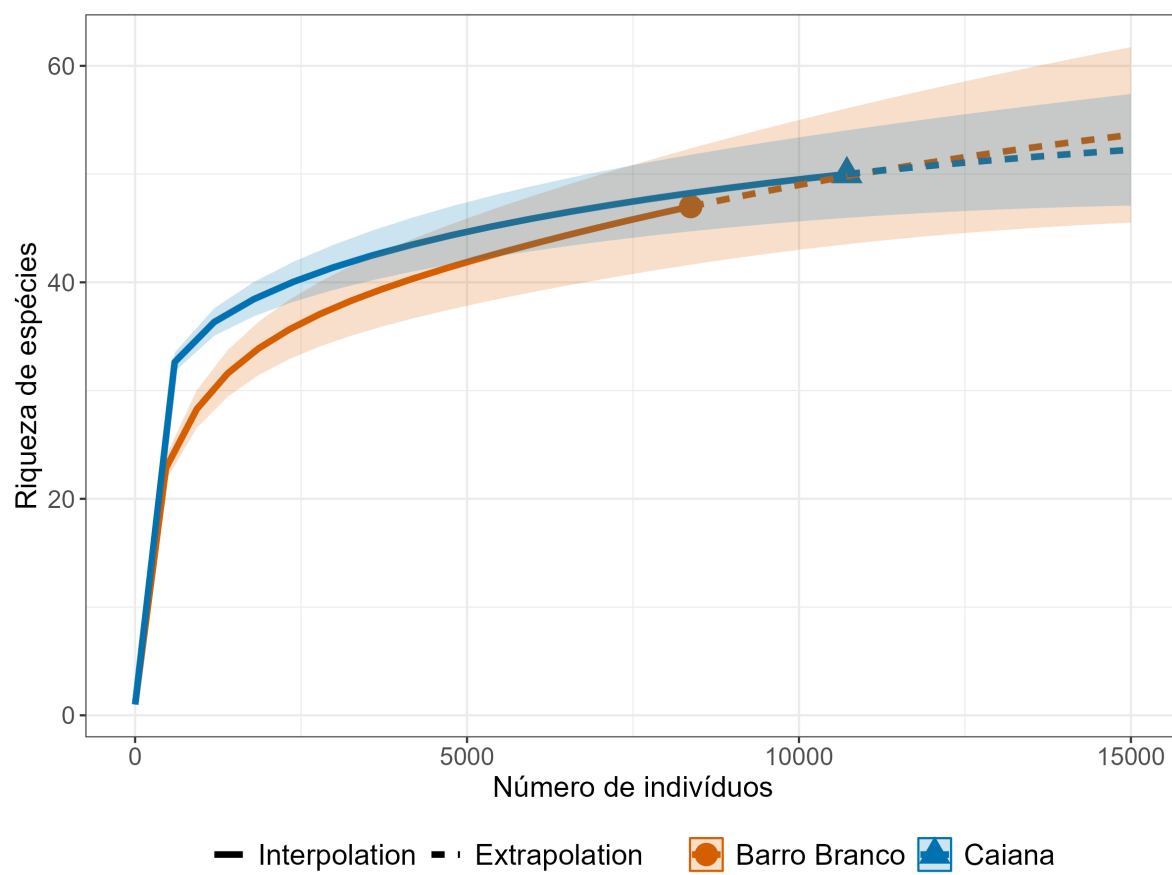


Figure 5

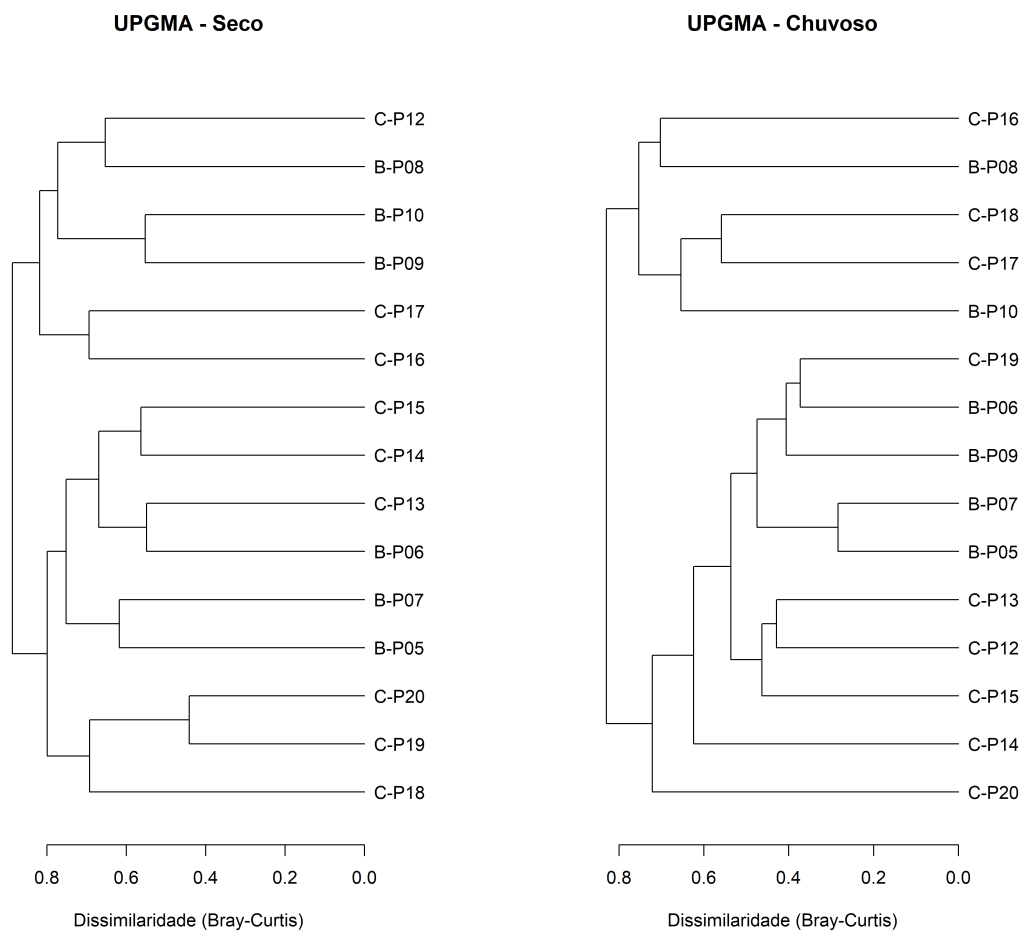


Figure 6

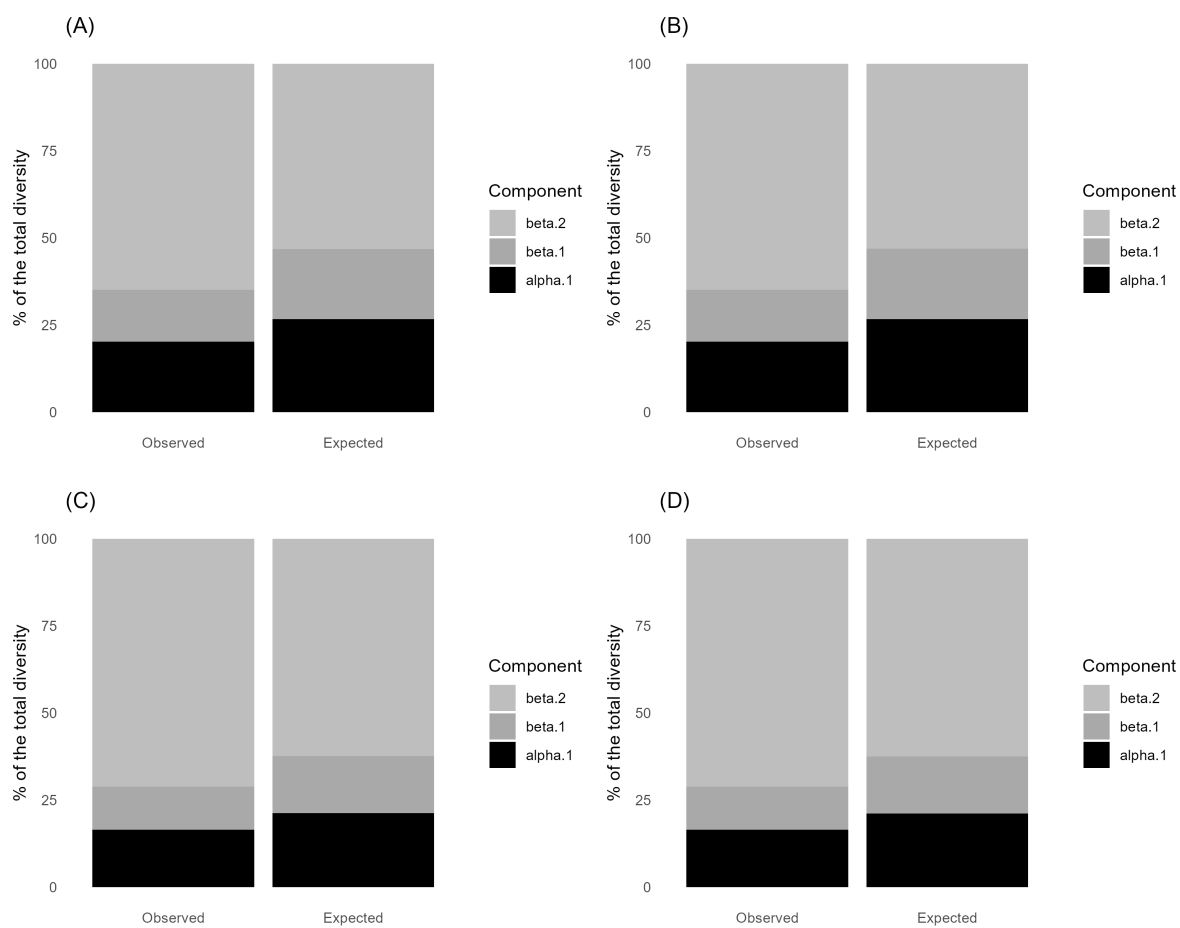


Figure 7

Table 1: Environment data GT table.

Variable	Ponto5	Ponto6	Ponto7	Ponto8
w.Cond.uS.cm	137.5 (119.2-164.4)	148 (134.1-156.8)	137.5 (98.3-160.2)	154.2 (141.6-161.8)
w.Turb.ntu	5 (2-10.7)	4.9 (2.1-14.1)	4.9 (1.5-9.2)	5.7 (4-6.8)
w.pH	5.9 (4.7-7.5)	5.7 (4.6-7.7)	5.1 (4.5-6)	5.5 (5-6)
m.Elev.m	107.1 (107.1-107.1)	112.8 (112.8-112.8)	89 (89-89)	89.1 (89.1-89.1)
m.Vel.m.s	0.073 (0.025-0.161)	0.086 (0.035-0.24)	0.135 (0.022-0.207)	0.036 (0-0.115)
m.Slope	1.4 (1-2)	2.7 (2.5-3)	2.4 (1.5-3)	1.5 (1-2)
m.Depth.avg.cm	12.8 (9-14)	23.7 (16.3-31.7)	23.8 (16-41)	43.3 (28.3-51.3)
m.Depth.mar.cm	11.8 (3-16)	21.7 (8-30)	18.8 (7-42)	48.3 (16-87)
s.Mud	100 (100-100)	100 (100-100)	95 (90-100)	74.2 (5-100)
s.Sand	0 (0-0)	0 (0-0)	5 (0-10)	25.8 (0-95)
s.Gravel.f	0 (0-0)	0 (0-0)	0 (0-0)	0 (0-0)
s.Rocks	0 (0-0)	0 (0-0)	0 (0-0)	0 (0-0)
h.Macrophytes	5.8 (5-10)	10 (0-30)	0 (0-0)	17.9 (0-60)
h.Grass	3.3 (0-10)	8.3 (5-15)	6 (0-15)	27.1 (15-50)
h.Subm.veg	7.5 (5-15)	8.3 (5-15)	5.8 (0-10)	5.8 (0-10)
h.Overh.veg	68.3 (40-90)	70.8 (50-85)	76.7 (60-90)	31.7 (25-50)
h.Leaf.l	7.5 (5-10)	11 (1-25)	15.8 (5-30)	18.3 (10-30)
h.Root.masses	5 (0-15)	10.8 (5-15)	12.5 (5-20)	4.2 (0-10)
h.Algae	2.5 (0-5)	0.2 (0-1)	5.5 (0-21)	19.2 (15-25)
h.Debris	30.8 (20-45)	12.7 (5-20)	17.7 (5-30)	21.7 (5-35)

Table 2: Nutrient concentrations at the sampling sites.

Variable	Ponto5	Ponto6	Ponto7	Ponto8
n.Ammonia.NH3.ug.L	54.8 (39.3-64.3)	58.9 (47.5-77)	72 (33.4-115.7)	81.6 (68.9-93)
n.Nitrite.NO2.ug.L	0 (0-0)	0 (0-0)	0 (0-0)	0 (0-0)
n.Nitrate.NO3.ug.L	6 (0-9.7)	8.8 (6.9-12.6)	5.5 (0-12.6)	18.8 (11.1-31.1)
n.Orthophosphate.SRP.ug.L	0 (0-0)	0 (0-0)	1 (0-3)	0 (0-0)
n.Phosphorus.TP.ug.L	0 (0-0)	4.6 (0.3-10.3)	50.1 (0-150.3)	9 (0-27)

Table 3: Density (ind/m²) taxa GT table.

Stream	Barro Branco Stream	...3	...4	...5	...6	...7	...8
Season	Wet						Dr
Site	Ponto5	Ponto6	Ponto7	Ponto8	Ponto9	Ponto10	Po
Alo.dadayi	0 (0)	0 (0)	0 (0)	5.3 (6.6)	0 (0)	0.4 (0.3)	0 (0)
Alo.hamulata	0 (0)	0 (0)	0 (0)	1.1 (1.3)	0 (0)	0.1 (0.1)	0 (0)
Alonella.sp	0 (0)	0 (0)	0 (0)	0.4 (0.5)	0 (0)	0 (0.1)	0 (0)
Bdelloidea	1.7 (0.5)	1.3 (0)	1.5 (0.5)	3.4 (0.8)	2 (0.8)	2.9 (1.3)	0.6
Bra.patulus	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Cephalodella.sp	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0.1 (0.1)	0 (0)
Colurella.sp	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Copepodite	0.5 (0.4)	0.1 (0)	0.1 (0)	12.1 (14.5)	0 (0)	0.5 (0.8)	0 (0)
Chydorus.sp	0 (0)	0 (0)	0 (0)	3.2 (3.5)	0 (0)	0.2 (0.1)	0 (0)
Dicranophoroides.sp	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Dia.birgei	0 (0)	0 (0)	0 (0)	0.1 (0.1)	0 (0)	0.1 (0.1)	0 (0)
Dia.spinulosum	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Dis.aculeata	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Euc.dilatata	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Fil.longiseta	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Ily.spinifer	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0.1 (0.1)	0 (0)
Lec.aculeata	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lec.bulla	0 (0)	0 (0)	0 (0)	0 (0.1)	0 (0)	0 (0)	0 (0)
Lec.cornuta	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lec.crepida	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lec.curvicornis	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lec.furcata	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lecane.sp1	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lec.leontina	0 (0)	0 (0)	0 (0)	0.5 (0.5)	0 (0)	0 (0.1)	0 (0)
Lec.lunaris	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lec.luna	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lec.monostyla	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lecane.sp2	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lec.papuana	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lec.quadridentata	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lecane.sp3	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Leydigia.sp	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lepadella.sp1	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lepadella.sp2	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lep.ovalis	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Lep.patella	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Mesocyclops.sp	0 (0)	0 (0)	0 (0)	0.6 (0.6)	0 (0)	0 (0)	0 (0)
Macrocylops.sp	0 (0)	0 (0)	0 (0)	0.2 (0.2)	0 (0)	0 (0)	0 (0)
Macrochaetus.sp	0 (0)	0 (0)	0 (0)	0.2 (0.3)	0 (0)	0 (0)	0 (0)
Macrothrix.sp	0 (0)	74 0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Mac.mira	0 (0)	0 (0)	0 (0)	0.2 (0.3)	0 (0)	0 (0)	0 (0)
Mac.laticornis	0 (0)	0 (0)	0 (0)	0.1 (0.1)	0 (0)	0 (0)	0 (0)
Moi.minuta	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Myt.crassipes	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Mytilina.sp	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Myt.bisulcata	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
N.elli	0.2 (0.2)	0.1 (0.1)	0.1 (0)	2.2 (2.6)	0 (0.1)	0.2 (1.2)	0 (0)

Table 4: Partition data GT table.

Sistema	Período	Componente	Obs (%)	Esp (%)	SES	p-valor
Riacho Caiana (CA)	Seco	1 (micro-habitat)	16.40%	21.17%	-12.27	< 0.00
Riacho Caiana (CA)	Seco	2 (ponto)	28.84%	37.52%	-9.32	< 0.00
Riacho Caiana (CA)	Seco	(Riqueza total)	100.00%	100.00%	0.00	1.000
Riacho Caiana (CA)	Seco	1 (entre micro-habitats)	12.43%	16.35%	-4.84	< 0.00
Riacho Caiana (CA)	Seco	2 (entre pontos)	71.16%	62.48%	9.32	< 0.00
Riacho Caiana (CA)	Chuvoso	1 (micro-habitat)	16.40%	21.18%	-12.12	< 0.00
Riacho Caiana (CA)	Chuvoso	2 (ponto)	28.84%	37.52%	-9.29	< 0.00
Riacho Caiana (CA)	Chuvoso	(Riqueza total)	100.00%	100.00%	0.00	1.000
Riacho Caiana (CA)	Chuvoso	1 (entre micro-habitats)	12.43%	16.33%	-4.70	< 0.00
Riacho Caiana (CA)	Chuvoso	2 (entre pontos)	71.16%	62.48%	9.29	< 0.00
Riacho Barro Branco (BB)	Seco	1 (micro-habitat)	20.23%	26.63%	-11.64	< 0.00
Riacho Barro Branco (BB)	Seco	2 (ponto)	35.04%	46.86%	-9.03	< 0.00
Riacho Barro Branco (BB)	Seco	(Riqueza total)	100.00%	100.00%	0.00	1.000
Riacho Barro Branco (BB)	Seco	1 (entre micro-habitats)	14.81%	20.23%	-4.66	< 0.00
Riacho Barro Branco (BB)	Seco	2 (entre pontos)	64.96%	53.14%	9.03	< 0.00
Riacho Barro Branco (BB)	Chuvoso	1 (micro-habitat)	20.23%	26.64%	-11.86	< 0.00
Riacho Barro Branco (BB)	Chuvoso	2 (ponto)	35.04%	46.88%	-9.39	< 0.00
Riacho Barro Branco (BB)	Chuvoso	(Riqueza total)	100.00%	100.00%	0.00	1.000
Riacho Barro Branco (BB)	Chuvoso	1 (entre micro-habitats)	14.81%	20.24%	-4.88	< 0.00
Riacho Barro Branco (BB)	Chuvoso	2 (entre pontos)	64.96%	53.12%	9.39	< 0.00
Partição Total (PT)	Seco	1 (micro-habitat)	12.45%	19.41%	-21.58	< 0.00
Partição Total (PT)	Seco	2 (ponto)	22.18%	34.41%	-16.50	< 0.00
Partição Total (PT)	Seco	3 (riacho)	73.28%	84.12%	-5.60	< 0.00
Partição Total (PT)	Seco	(Riqueza total)	100.00%	100.00%	0.00	1.000
Partição Total (PT)	Seco	1 (entre micro-habitats)	9.73%	15.00%	-8.68	< 0.00
Partição Total (PT)	Seco	2 (entre pontos)	51.09%	49.72%	0.69	0.499
Partição Total (PT)	Seco	3 (entre riachos)	26.72%	15.88%	5.60	< 0.00
Partição Total (PT)	Chuvoso	1 (micro-habitat)	23.72%	51.72%	-45.73	< 0.00
Partição Total (PT)	Chuvoso	2 (ponto)	35.50%	69.45%	-35.41	< 0.00
Partição Total (PT)	Chuvoso	3 (riacho)	83.75%	96.48%	-9.42	< 0.00
Partição Total (PT)	Chuvoso	(Riqueza total)	100.00%	100.00%	0.00	1.000
Partição Total (PT)	Chuvoso	1 (entre micro-habitats)	11.78%	17.73%	-7.46	< 0.00
Partição Total (PT)	Chuvoso	2 (entre pontos)	48.25%	27.04%	13.64	< 0.00
Partição Total (PT)	Chuvoso	3 (entre riachos)	16.25%	3.52%	9.42	< 0.00

Apendices

Non-used figures and tables

 Non-used codes

1 `table(1:10)`